

ZHUYUAN LIU PORTFOLIO

MArch · ARB/RIBA Part 2
AA School of Architecture

Real-Time Visualisation · Immersive Media · Interactive Systems
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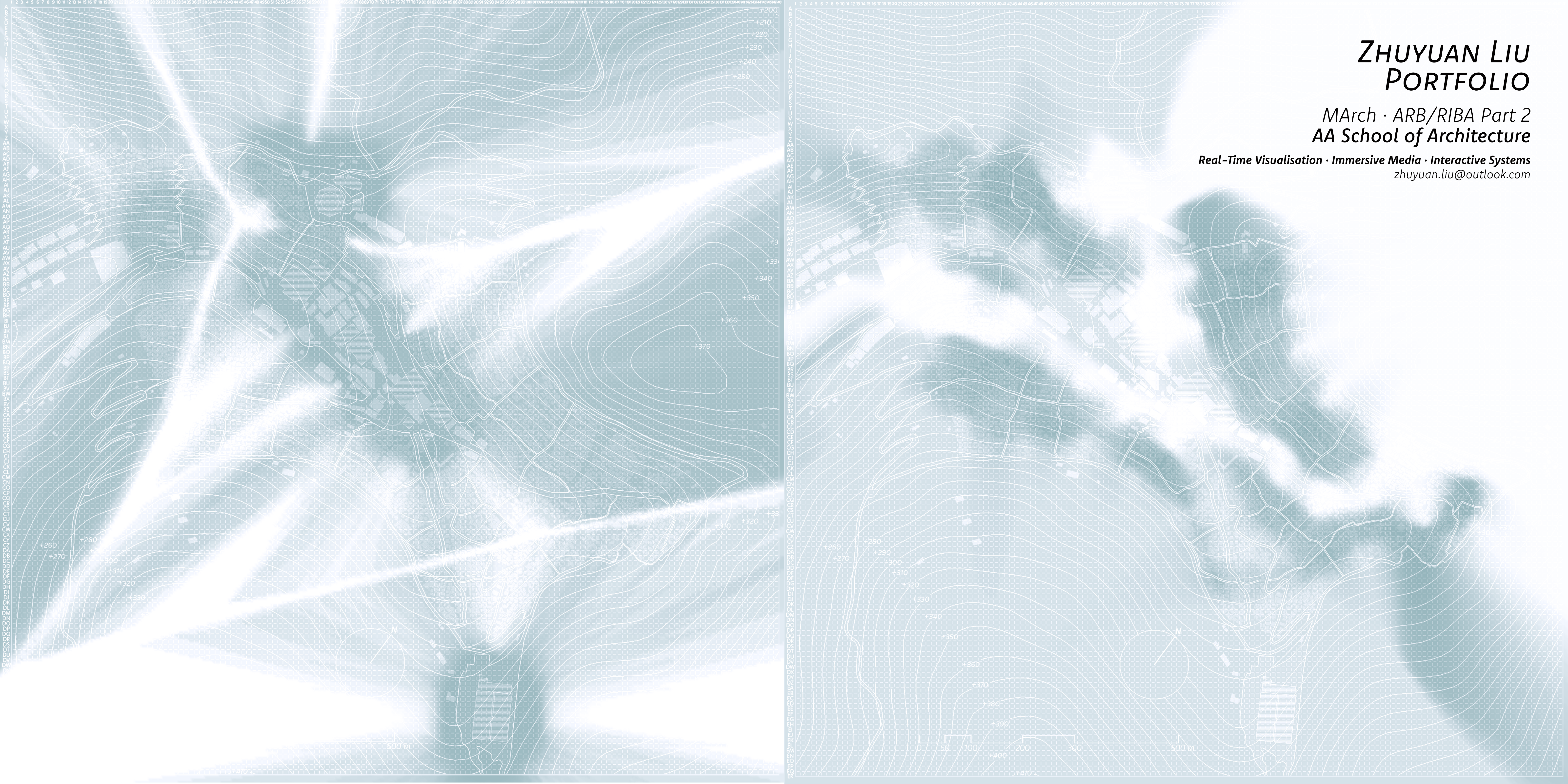


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1. SACRED INTELLIGENCE

TIBETAN BUDDHIST SPACES
BETWEEN TRADITION AND TECHNOLOGY

UNREAL ENGINE · RESPONSIVE MEDIA ·
PHYSICAL COMPUTING

Sacred Intelligence explores how emerging technologies can be integrated into Tibetan Buddhist learning spaces without overwhelming their cultural and ritual character. The project combines architectural design, real-time spatial visualisation, responsive media and physical prototyping to test how digital systems might support embodied experience rather than operate as a separate technological layer.

The assembly hall study explores how responsive media, material character, lighting and existing ritual objects could coexist within a single spatial environment while retaining the identity of the traditional interior.



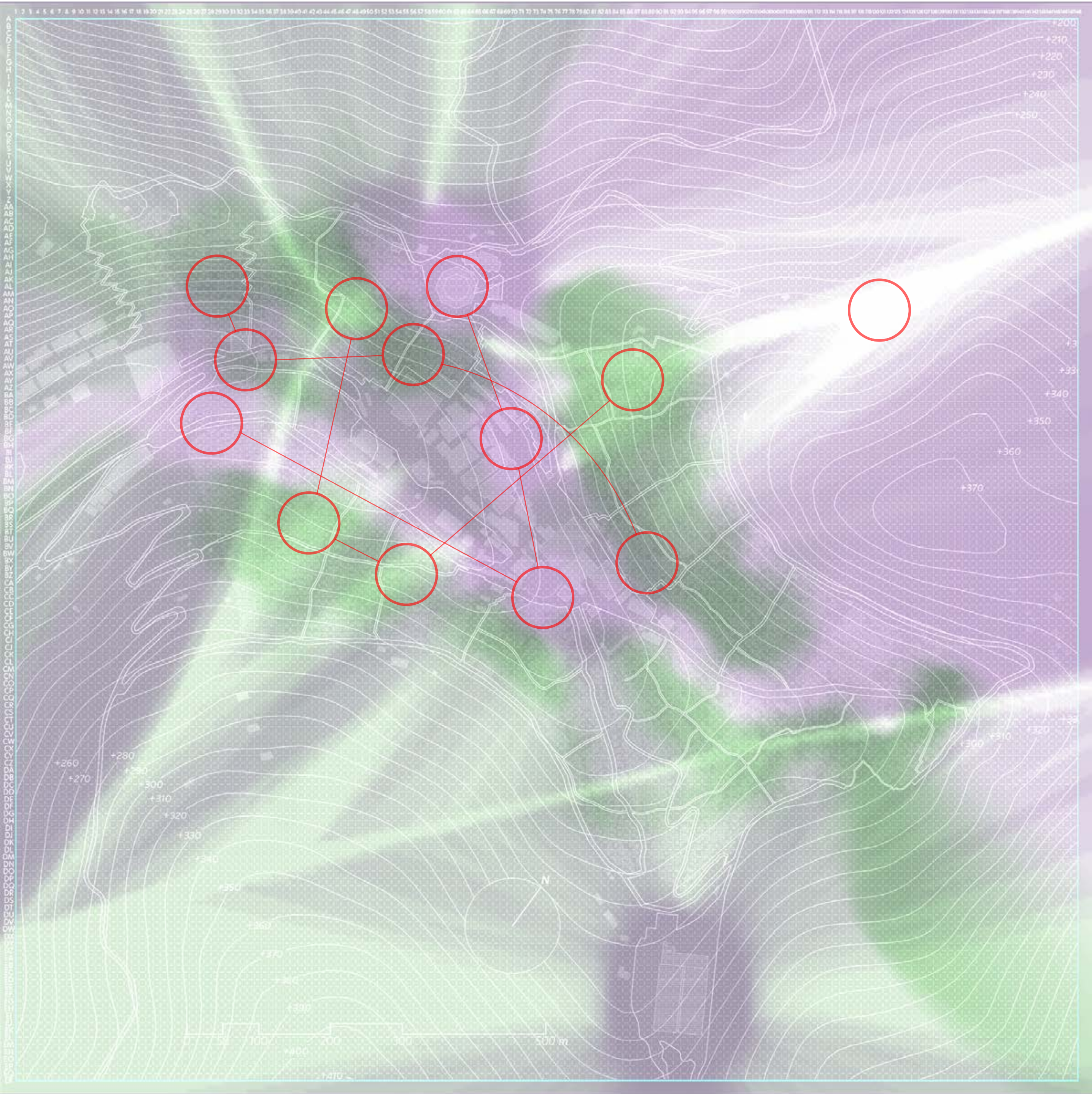
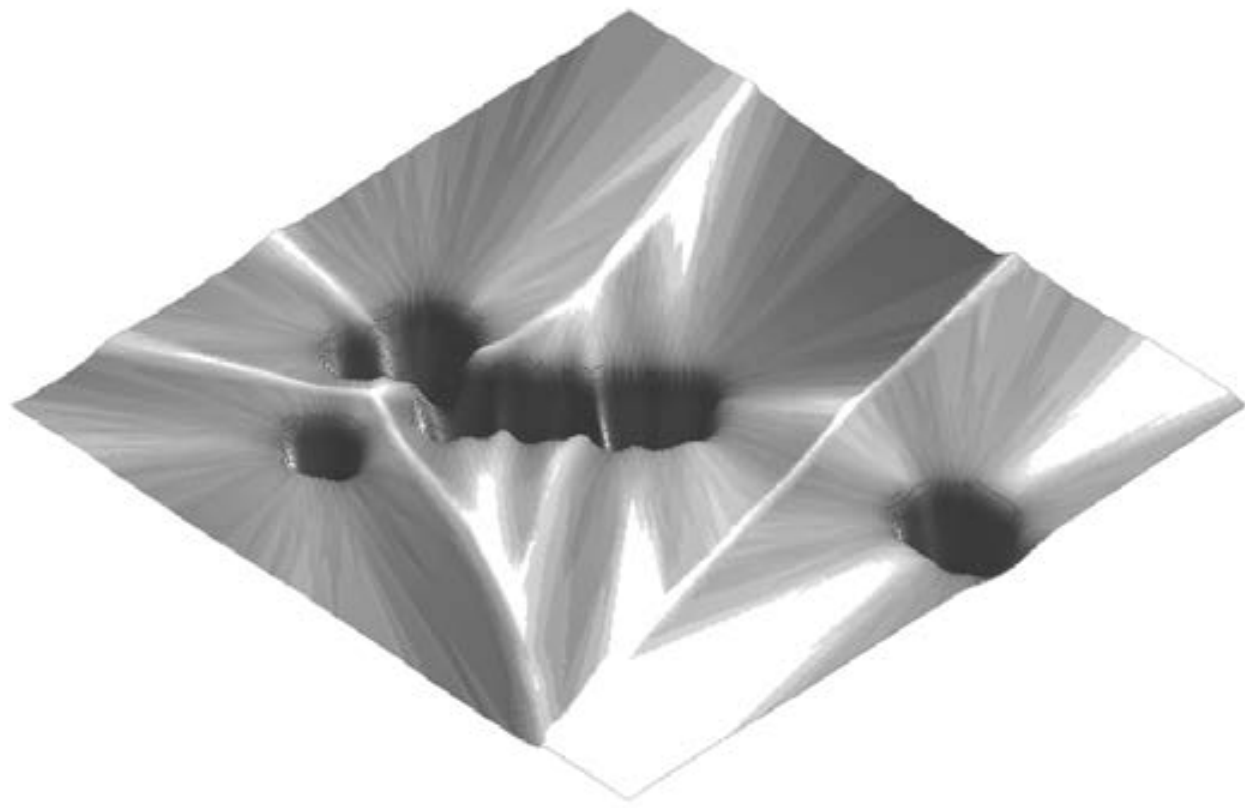
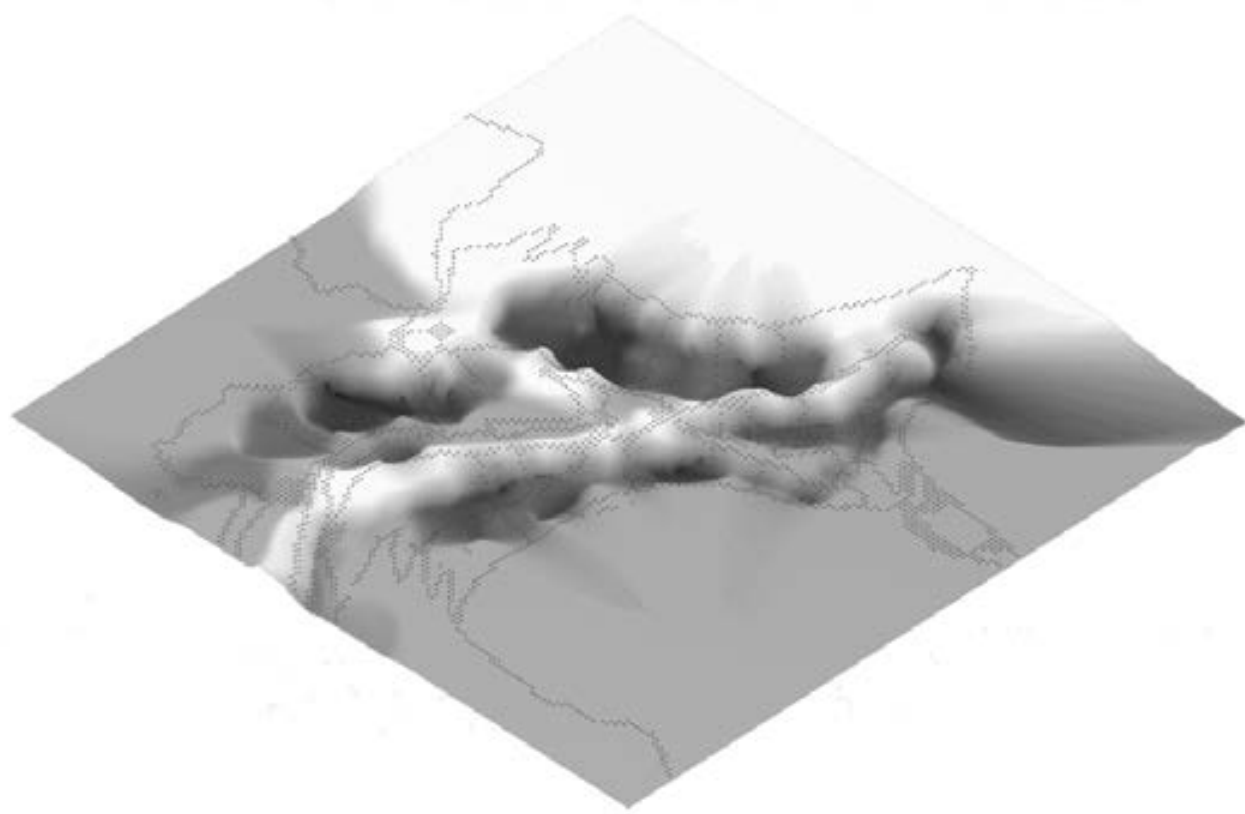
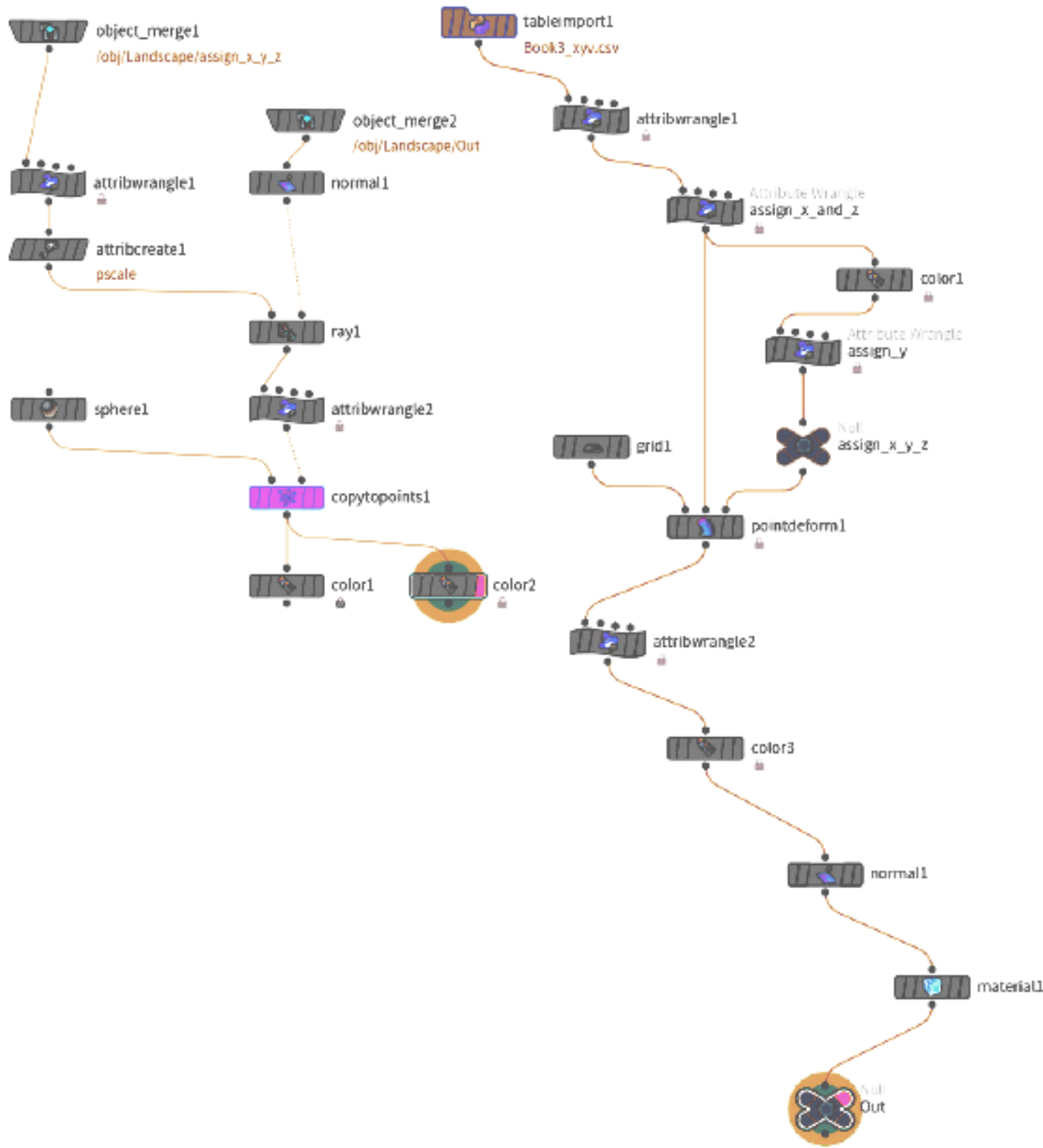
*A Mediated Assembly Hall
Technological Speculation on Tibetan Buddhist Learning Space*

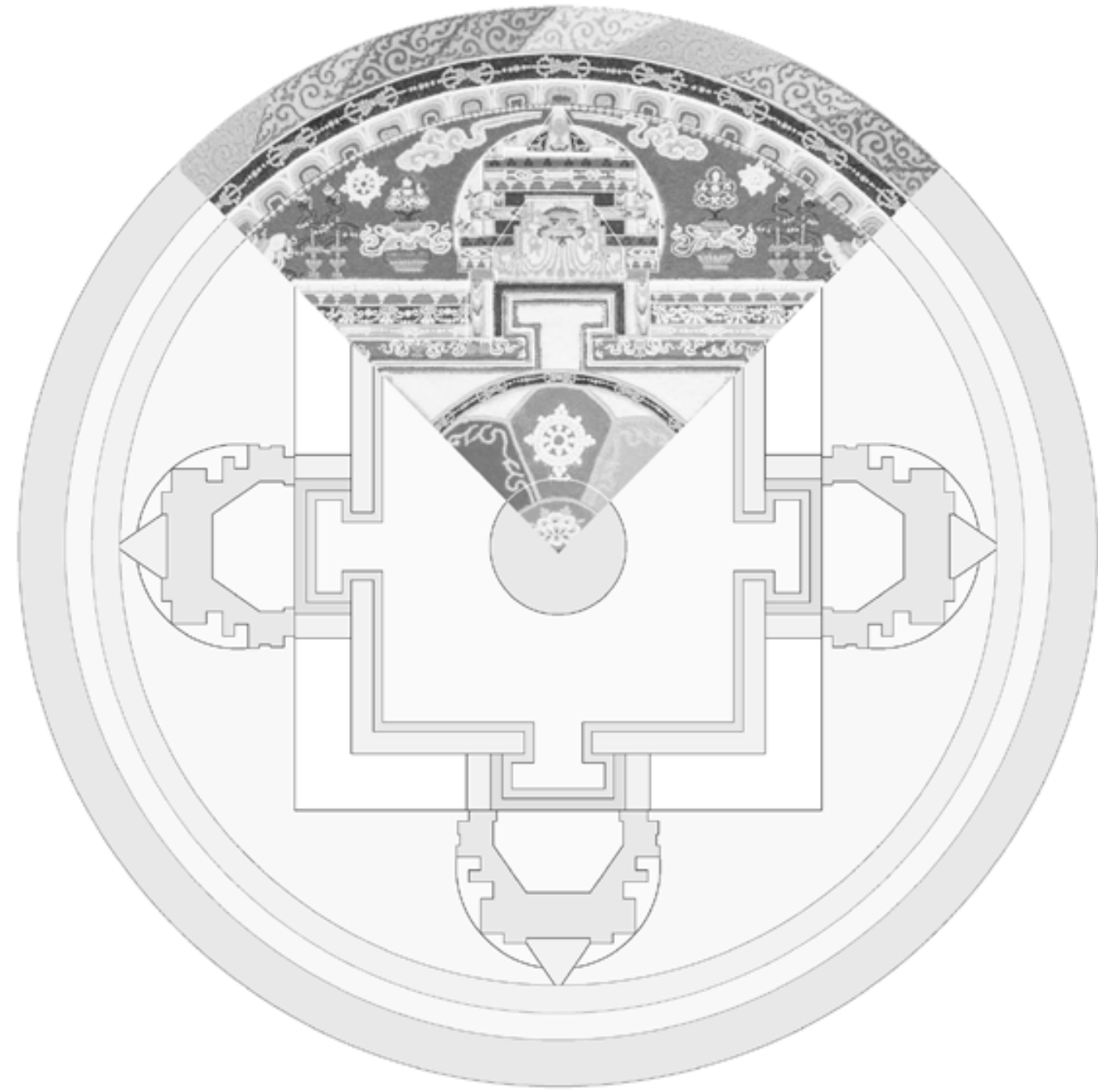
Computational Spatial Visualisation

The project uses procedural visualisation to make competing spatial forces legible. Religious attraction and administrative constraints are encoded as weighted datasets, generating overlapping fields that reveal areas of convergence, resistance and potential intervention.

Developed through a Houdini-based procedural workflow, the datasets are translated into three-dimensional fields and combined into a composite decision landscape, turning relationships that are difficult to read through conventional mapping into explicit visual information.

- REGIONS OF INTEREST TO **BOTH SIDES**
- AREAS OF INTEREST ONLY TO **BUDDHISTS**
- REGIONS PREFERRED ONLY BY **ADMINISTRATIVE FORCES**
- REGIONS OF INTEREST TO **NEITHER SIDE**





Translating Knowledge into Spatial Form

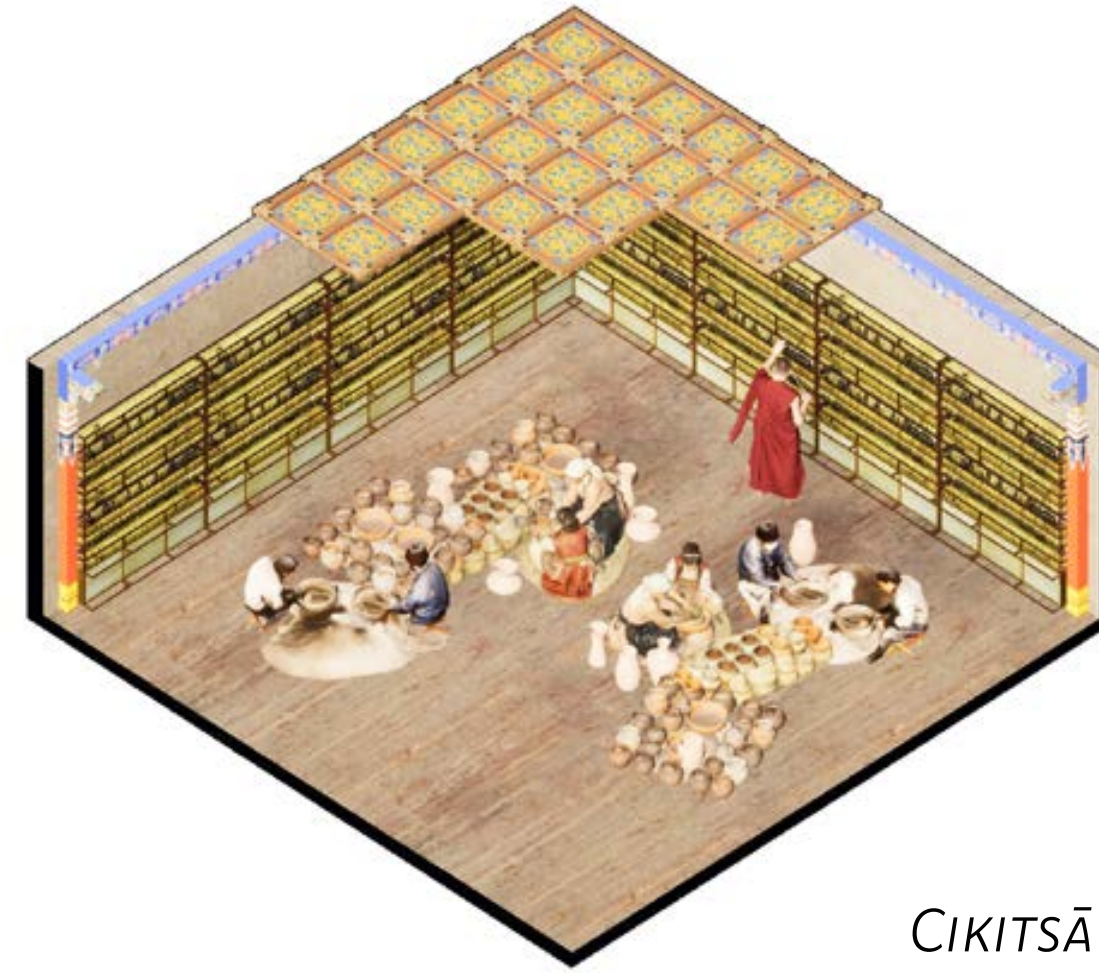
Five spatial studies developed and visualised in Unreal Engine

The Five Sciences provide five distinct conceptual starting points for spatial development. Medicine, language, arts and crafts, inner science and logic were translated into different forms of gathering, teaching, ritual and exchange.

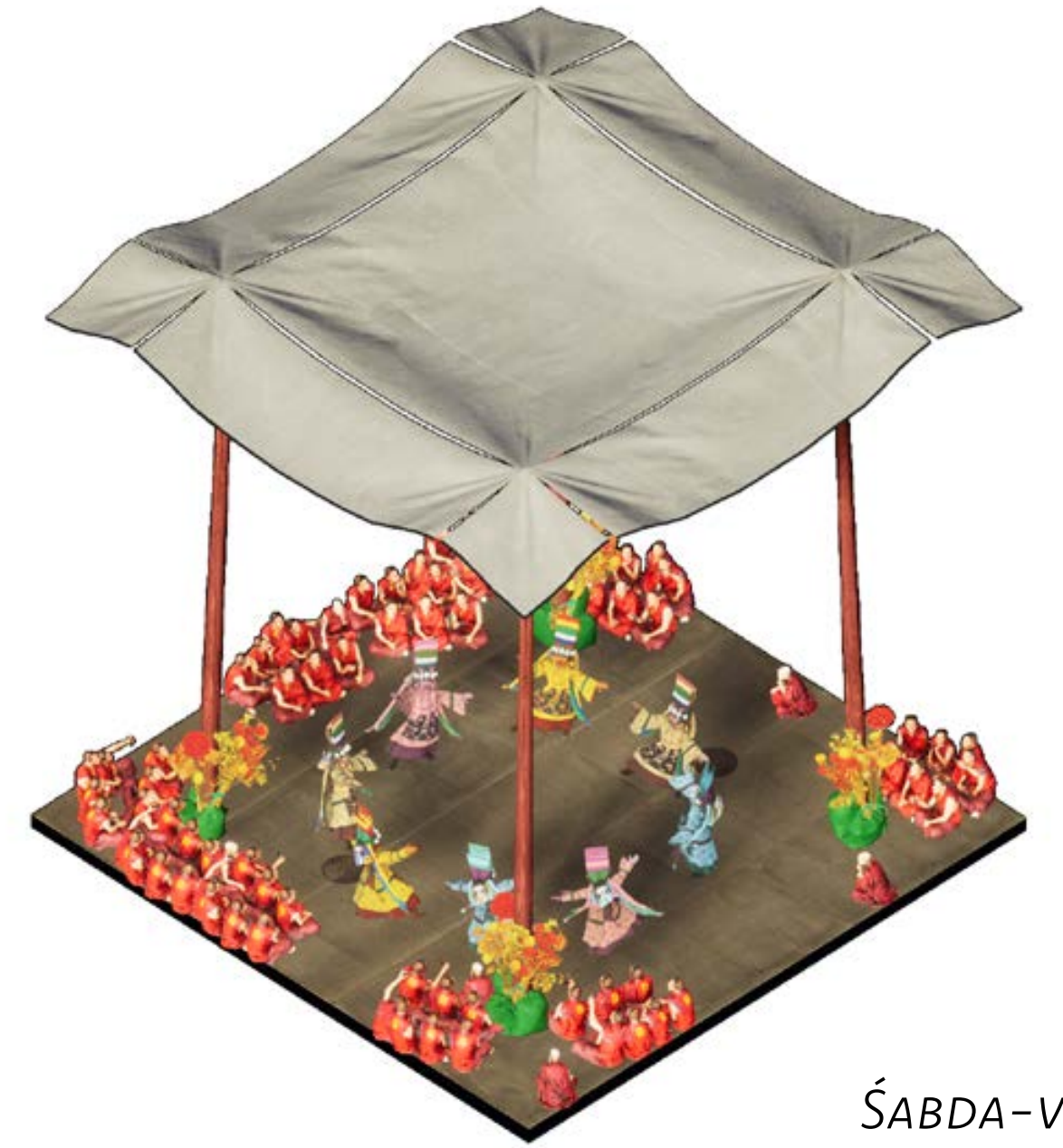
Each was developed as a distinct Unreal Engine environment, using spatial composition, material, lighting and atmosphere to test how abstract forms of knowledge could acquire different visual and experiential identities. The sand mandala was then selected for deeper development as both a spatial practice and a highly structured geometric system.



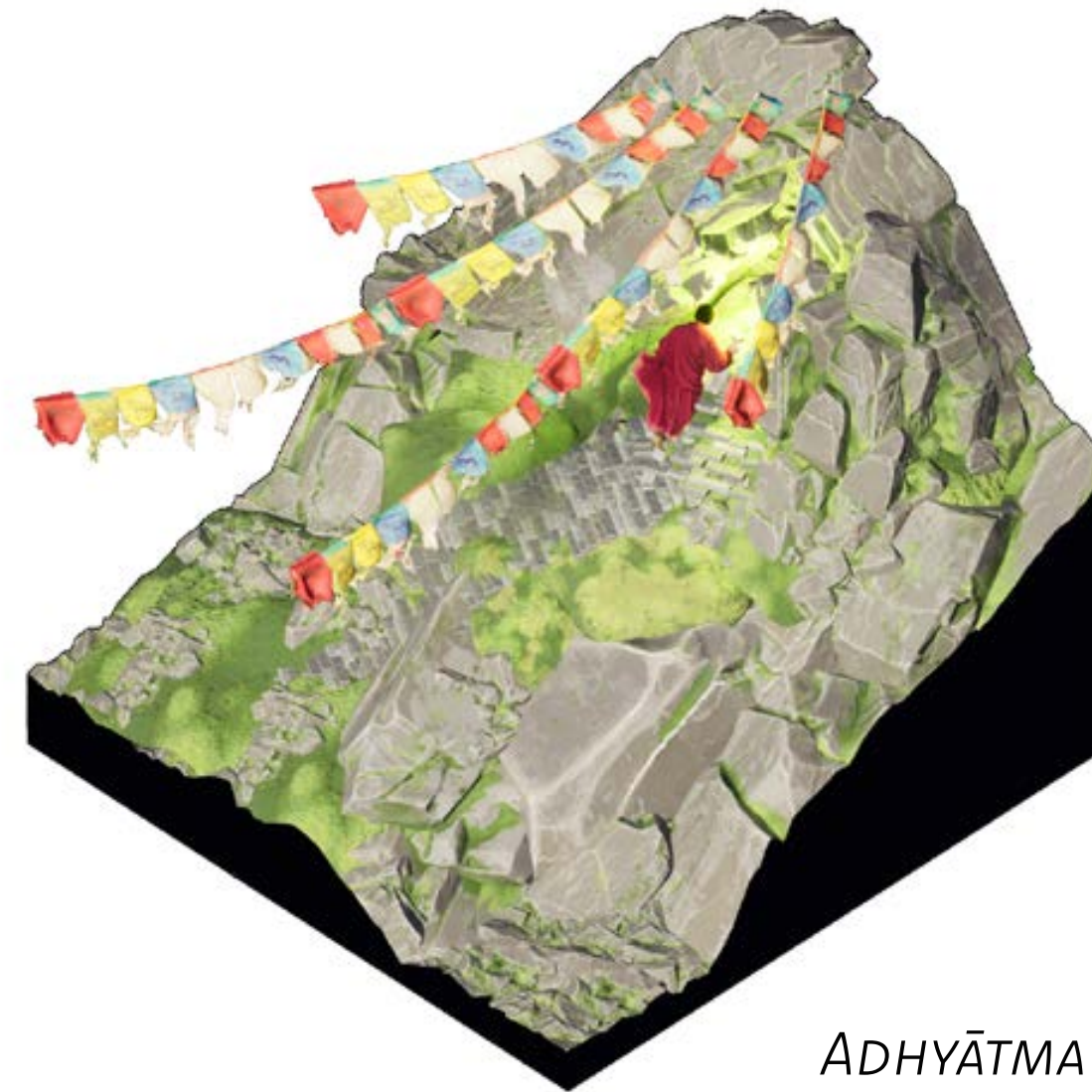
ŚILPA-VIDYĀ
"SCIENCE OF ARTS AND CRAFTS"



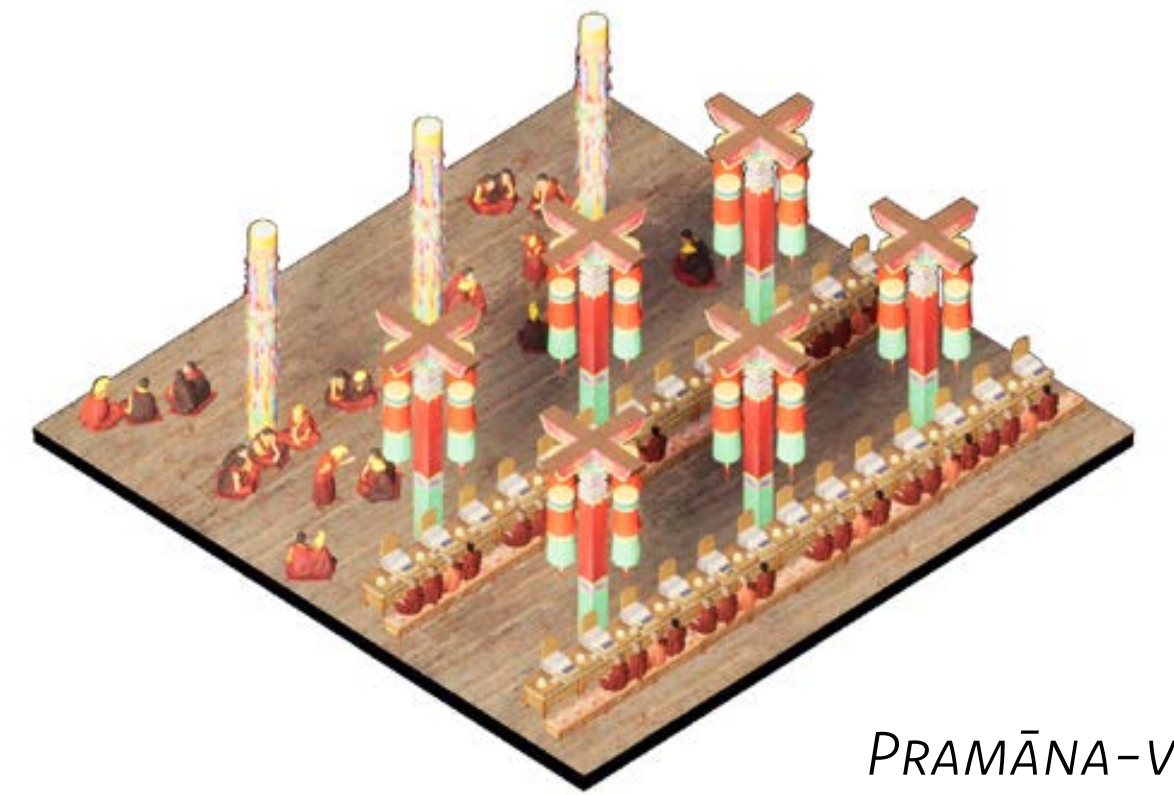
CIKITSĀ-VIDYĀ
"SCIENCE OF MEDICINE"



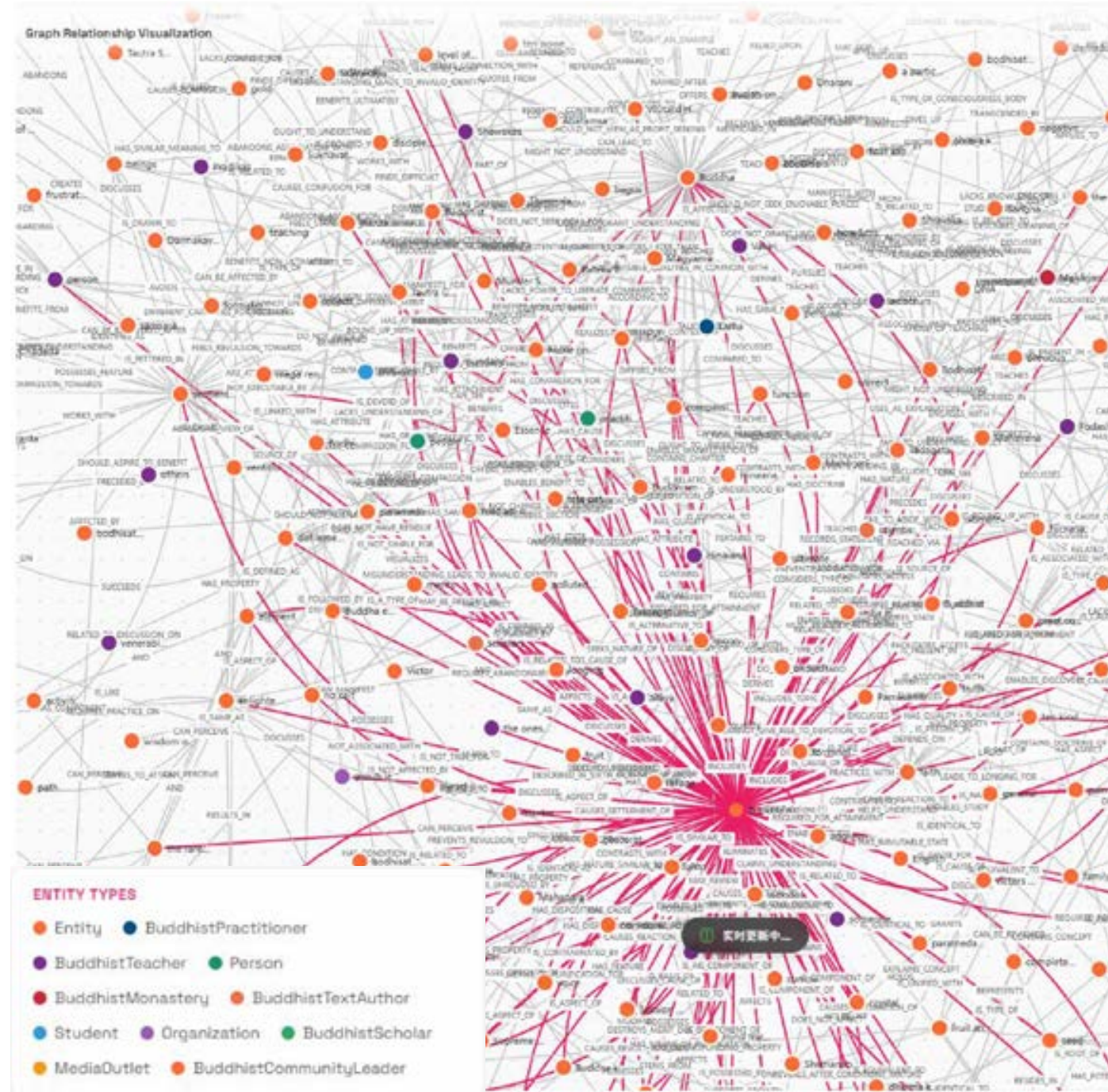
ŚABDA-VIDYĀ
"SCIENCE OF LANGUAGE"



ADHYĀTMA-VIDYĀ
"INNER SCIENCE"



PRAMĀNA-VIDYĀ
"SCIENCE OF LOGIC"



Zep-Based Knowledge Analysis

AI Pipeline

A Zep-based knowledge archive structures and retrieves relationships across the source material, while a custom Python application coordinates contextual reasoning, visual-state selection, generative media, post-processing and live playback.

Together, these components form a real-time media pipeline in which changing input drives continuously updated visual output.

Custom Python-based AI agent and media pipeline

- display_player
 - __pycache__
 - _init_.py
 - play_history_then_live.py
 - watch_and_postprocess.py
 - watch_display_loop.py
 - input_gateway
 - __pycache__
 - audio
 - clients
 - input_logs
 - server.py
 - settings.py
 - storage.py
 - transcribe_whisper.py
 - pipeline
 - __pycache__
 - outputs
 - run_live_pipeline.py
 - test_latest_pipeline.py
 - reasoner
 - __pycache__
 - image_selector.py
 - packet_reader.py
 - test_directed_selection.py
 - test_latest_packet.py
 - test_select_image.py
 - visual_state_reasoner.py
 - runtime_logs
 - display_loop.log
 - input_gateway.log
 - live_pipeline.log
 - s3_ambient.log
 - seedance_watcher.log
 - watch_and_postprocess.log
 - video_generation
 - __pycache__
 - keyframes
 - seedance_outputs
 - state
 - .env
 - 000_start.png
 - postprocess_latest_video.py
 - seedance_from_pipeline.py
 - seedance_interactive.py
 - watch_seedance_from_pipeline.py
 - visual_library
 - images
 - mural_index.json
 - .txt
 - system_launcher_final.py
- Manages live playback and updates
- Captures and transcribes input
- coordinates system logic and data flow
- retrieves context and selects visual states
- Monitors system execution
- Sends prompts to Seedance and post-processes outputs
- Organises indexed visual references and mural states
- Runtime Coordinator

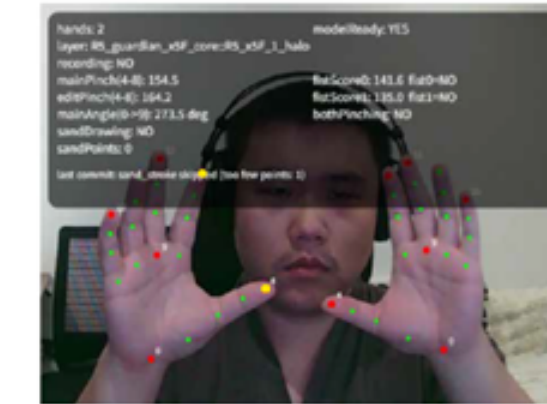
Real-Time Gesture Interaction

Hand landmarks are captured and translated into structured real-time inputs for downstream visual and spatial responses.

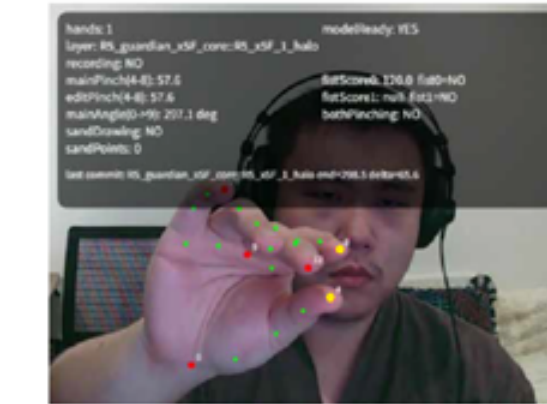
Both hands clenched (fists) → execute Draw All, triggering the full drawing sequence.



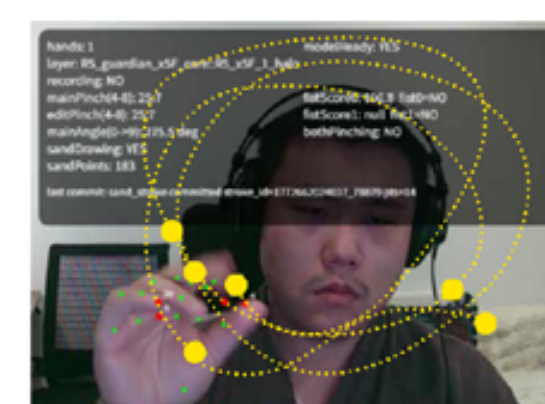
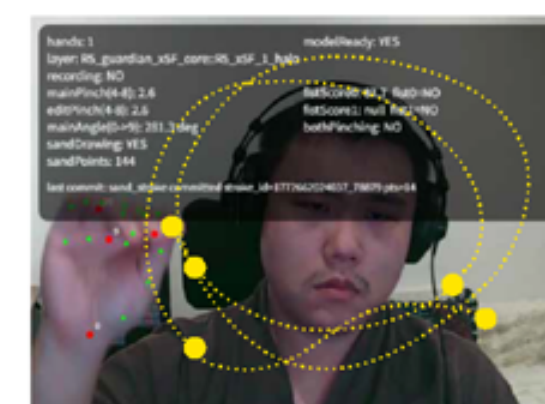
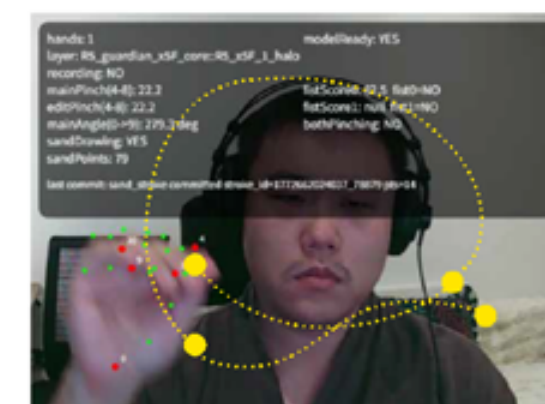
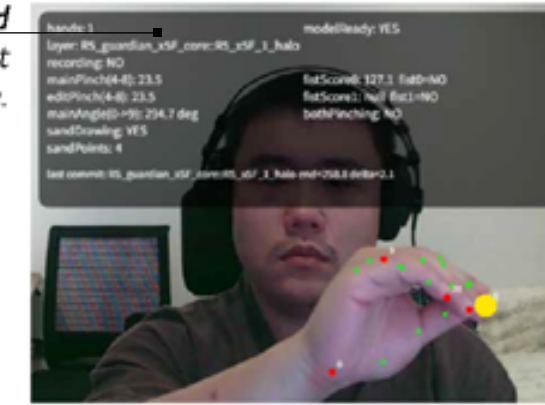
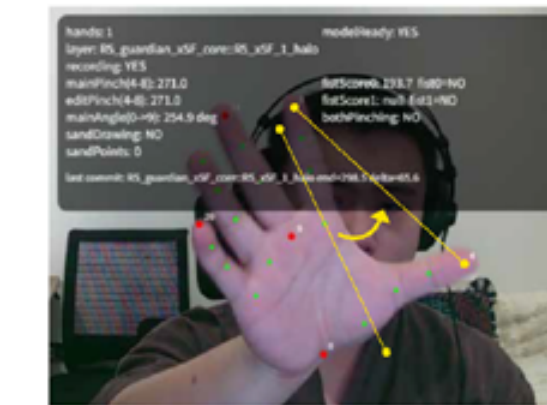
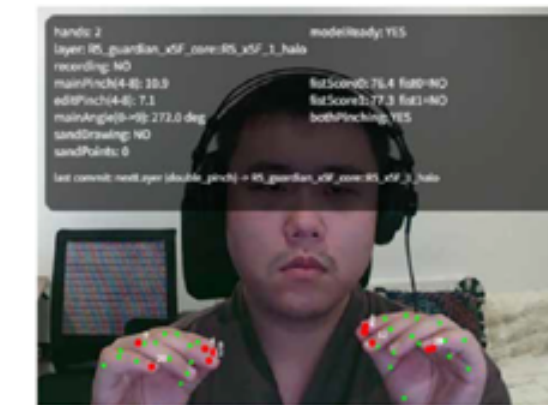
Double-hand rotation around an anchor point → edit stroke paths, enabling adjustments to existing trajectories.



Single-hand rotation around an anchor point → edit stroke paths, enabling adjustments to existing trajectories.



Both hands pinching → switch layers, allowing the user to move between different drawing layers.



DESIGN IN USE

CINEMATIC VISUALISATION &
1:1 PHYSICAL PROTOTYPING

HALL OF MAKING
CLOISTER OF EXCHANGE
HALL OF RESONANCE
HALL OF REVELATION

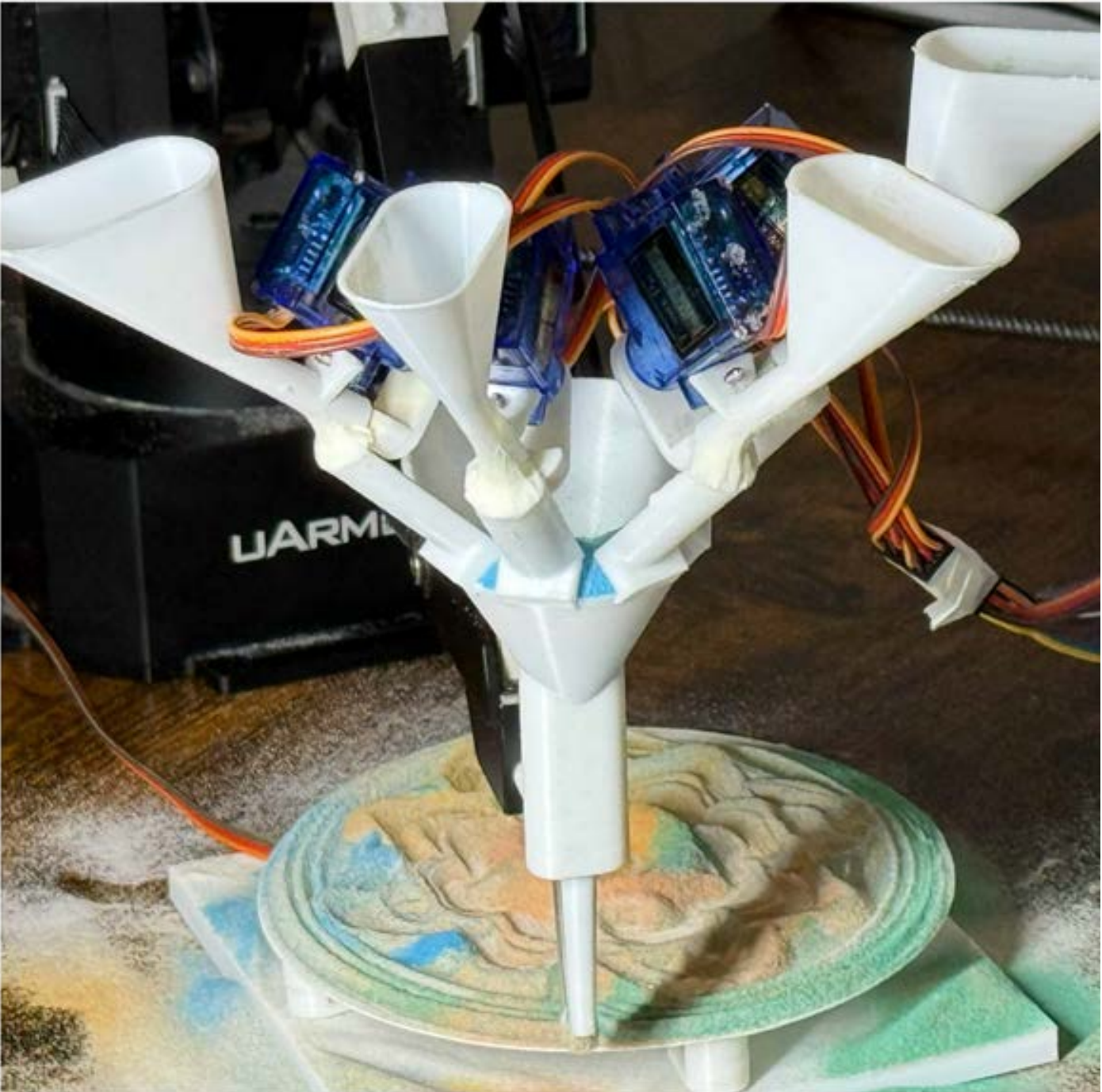
The project integrates robotic fabrication, responsive media and remote participation into four architectural spaces, each developed around a specific form of learning, ritual or collective practice. Key interactions were then tested through 1:1 prototypes, connecting spatial proposals with computational workflows, physical making and their intended cultural context.



Robotic Sand Mandala

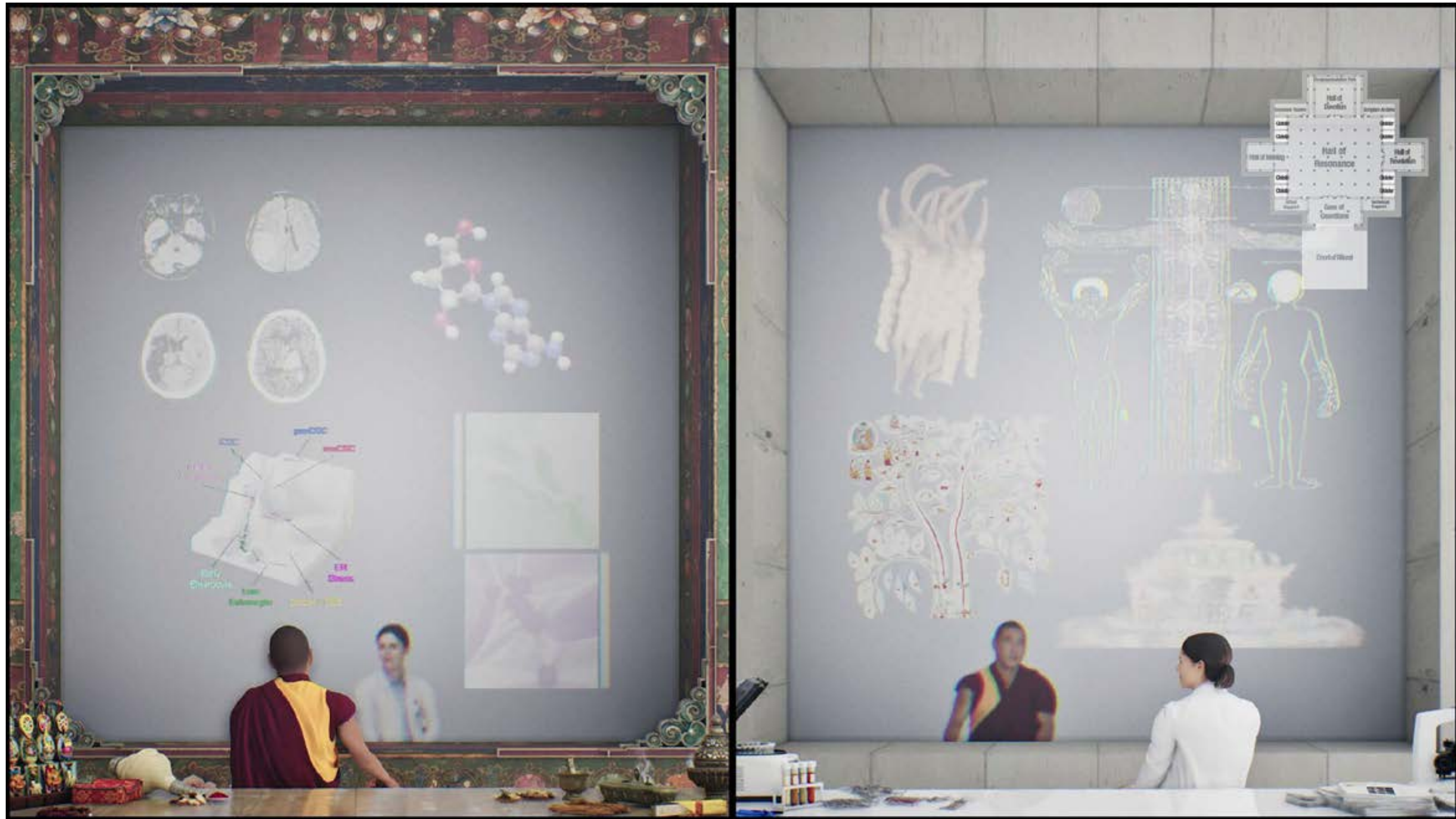
During the annual sand mandala ritual, a mandala master at Gonggar transmits drawing instructions to a robotic installation in London. Rather than observing a digital reproduction, remote participants can take part in the same process of making, extending the embodied ritual across geographical distance.

Scan / click
Prototype test video



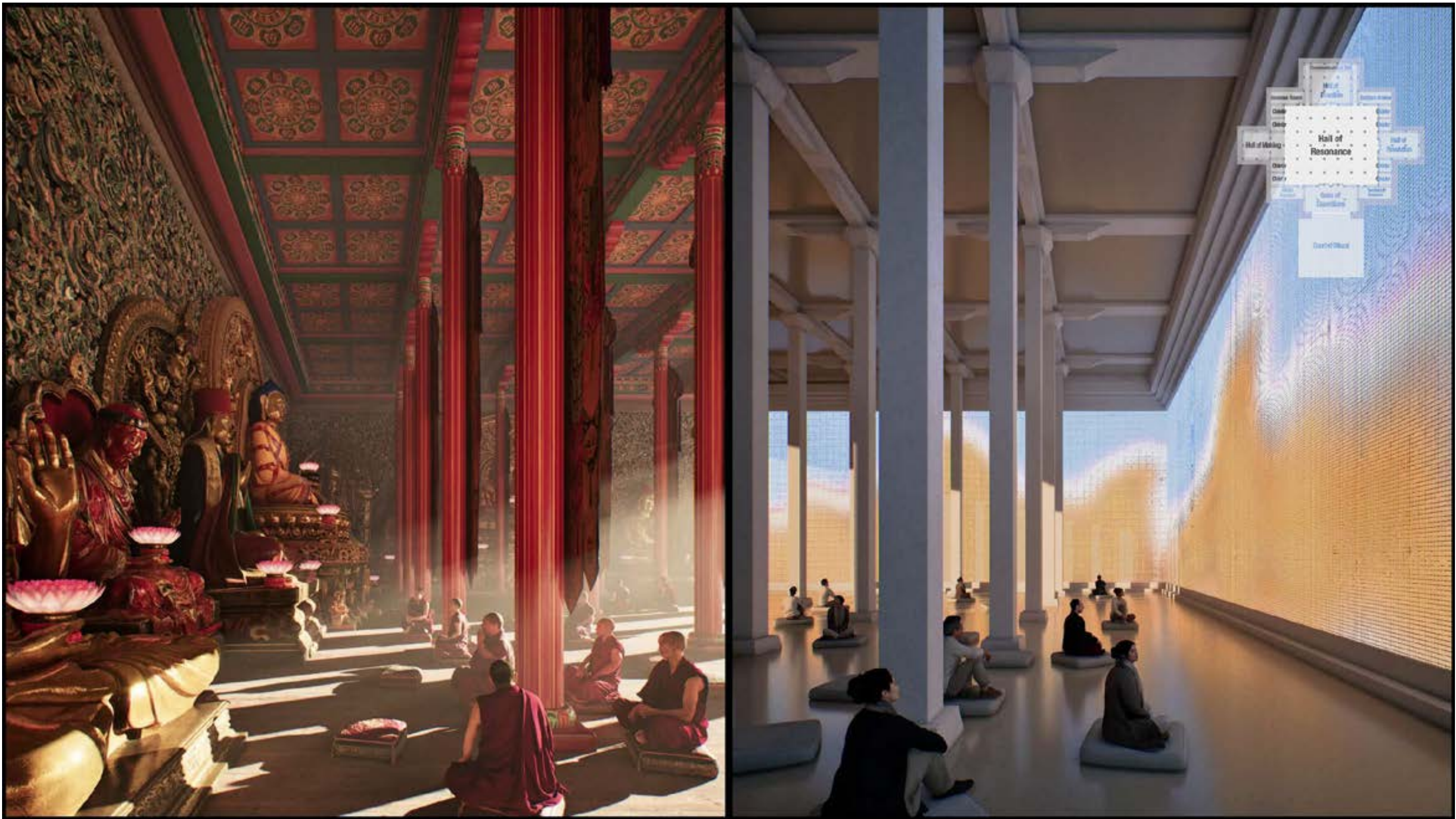
Multi-Colour Sand Mandala Fabrication System

The 1:1 prototype integrates four independent sand channels, enabling controlled colour switching during robotic drawing. Regulated outlets and a soft-brush tip stabilise material flow, demonstrating multi-colour sand deposition as a repeatable physical making process rather than a digital simulation.



Cloister of Exchange

The Cloister of Exchange connects distant monastic communities through a shared environment for teaching, discussion and study. The spatial study tests how remote participation, media surfaces and collective gathering can operate as a continuous experience rather than as a conventional video call.



Hall of Resonance

The Hall of Resonance explores collective meditation across distance. Breathing, presence and environmental conditions from one location are translated into subtle changes in light, atmosphere and responsive media in the other, communicating participation without relying on direct visual representation.

The project does not propose technology as a replacement for Buddhist practice. Instead, it tests where digital systems can meaningfully support learning, ritual and communication while preserving forms of practice that depend on direct human presence. The resulting architecture treats technology as an embedded spatial medium rather than an external layer.

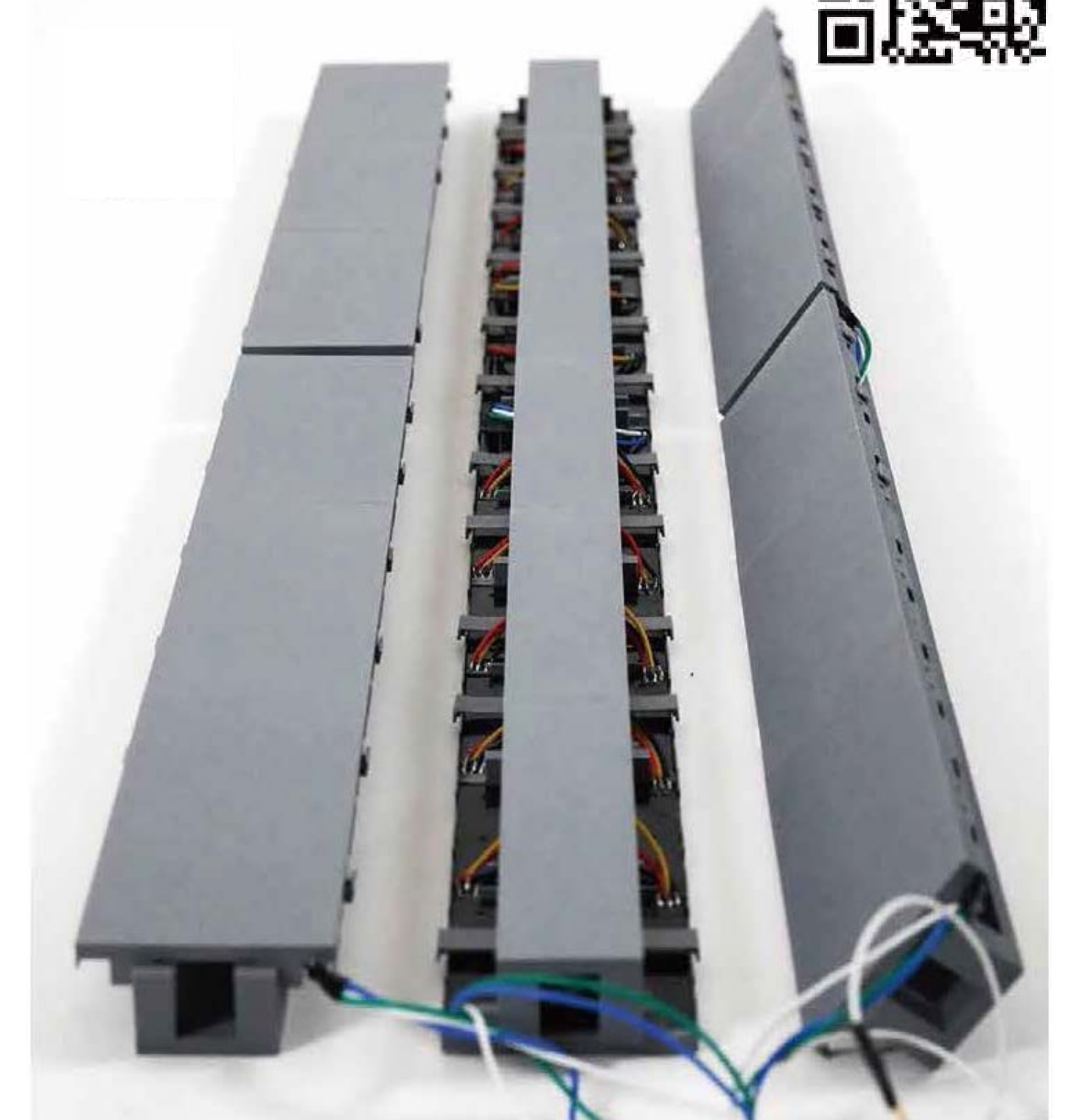


Hall of Revelation – Responsive Mural

Generative Media as a Responsive Spatial Surface

During daily chanting at Gonggar, spoken teachings and contextual inputs are processed through the AI system and translated into evolving mural imagery in London. Rather than displaying a fixed render or recording, the surface operates as a continuously changing visual environment driven by contextual input.

Scan / click
Generated Mural Archive



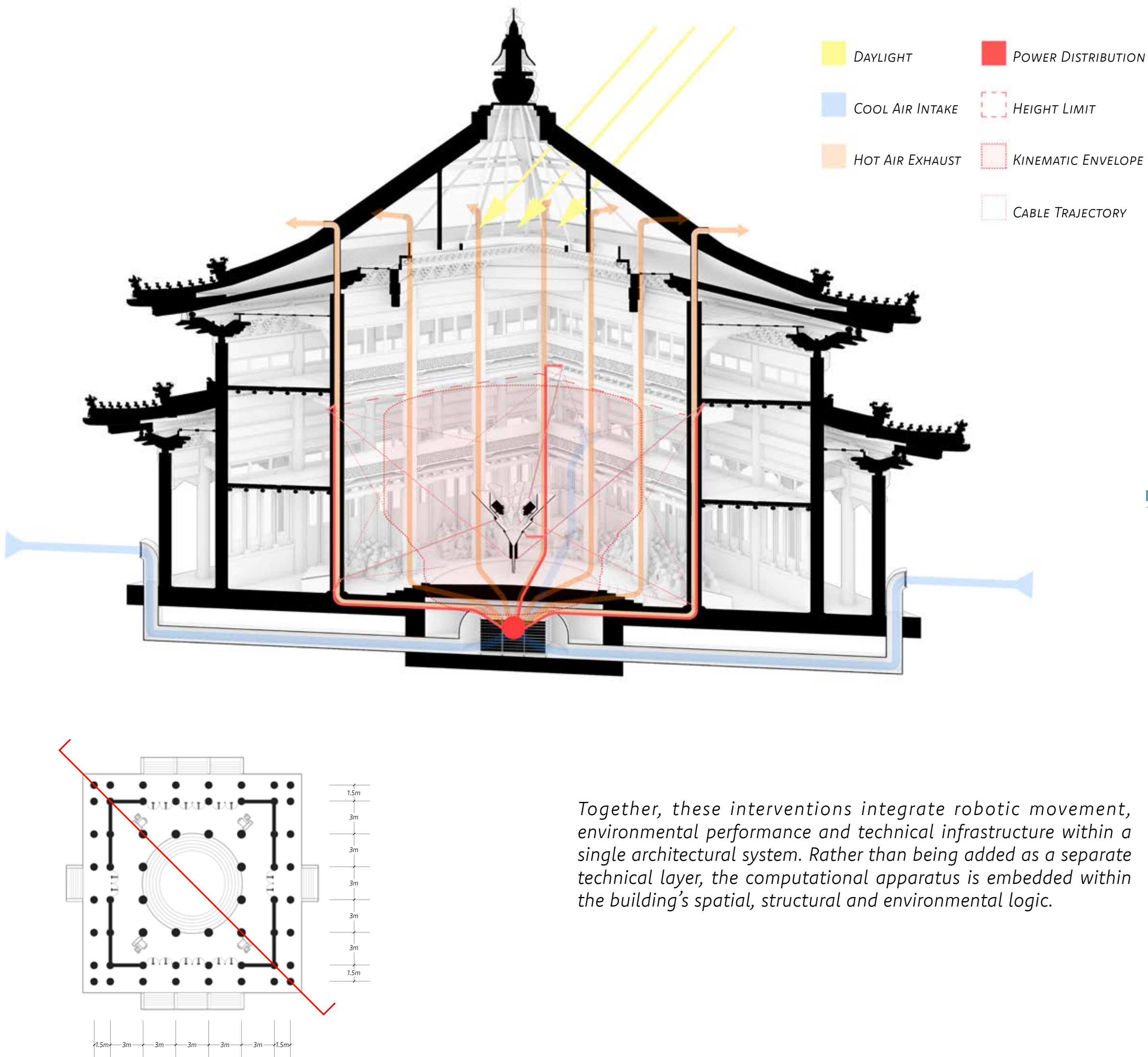
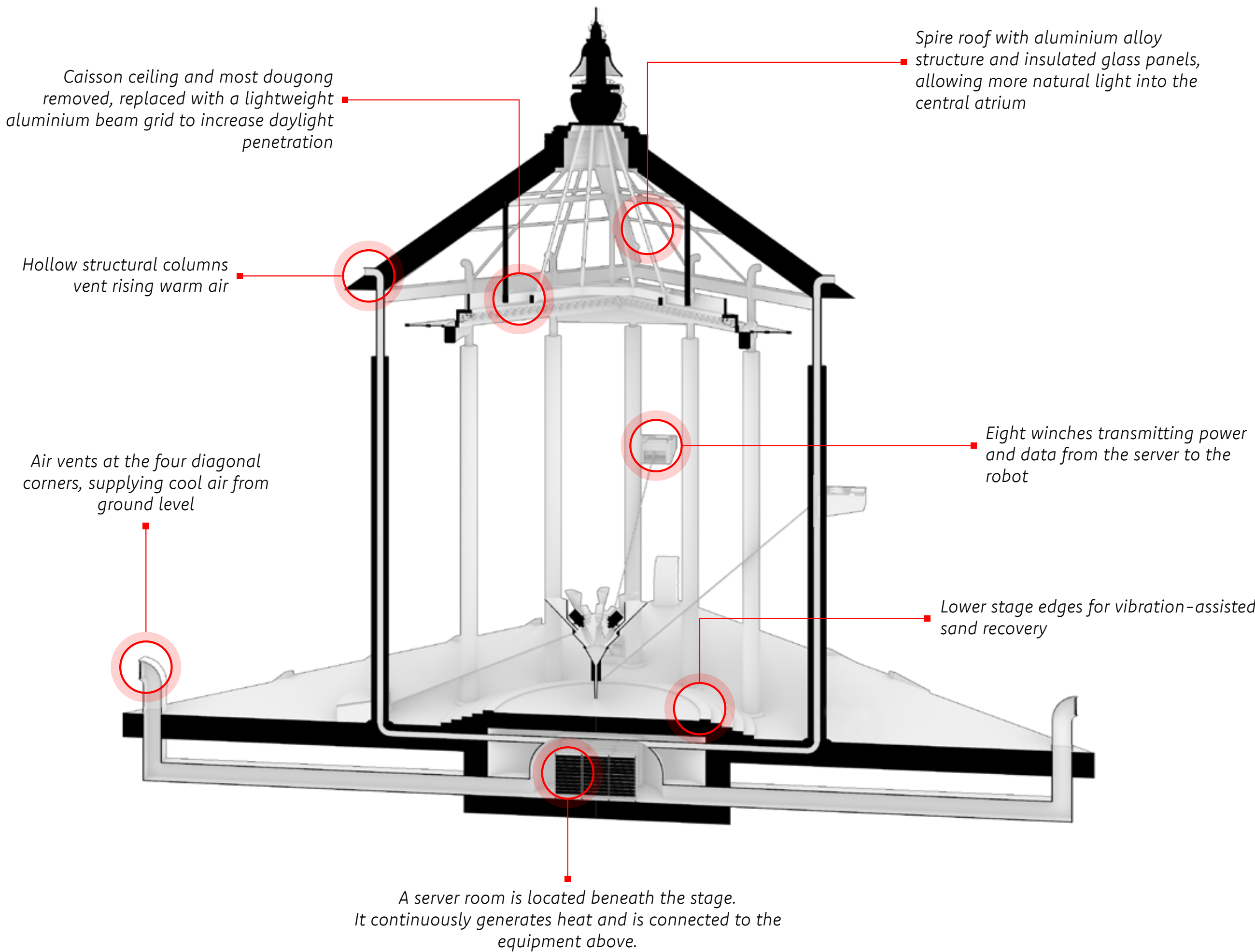
1:1 Responsive Media Prototype

The 1:1 prototype tests dynamically generated visual content as an integrated responsive media surface rather than a standalone screen. It demonstrates how responsive imagery can operate at spatial scale and become part of the surrounding environment.

Architectural System Integration

Infrastructure for Robotic Operation

The robotic mandala system is integrated through targeted architectural modifications. Roof alterations increase daylight penetration, hollow structural columns support passive heat exhaust, and dedicated air, power and data routes connect the server room to the robotic apparatus.



Together, these interventions integrate robotic movement, environmental performance and technical infrastructure within a single architectural system. Rather than being added as a separate technical layer, the computational apparatus is embedded within the building's spatial, structural and environmental logic.

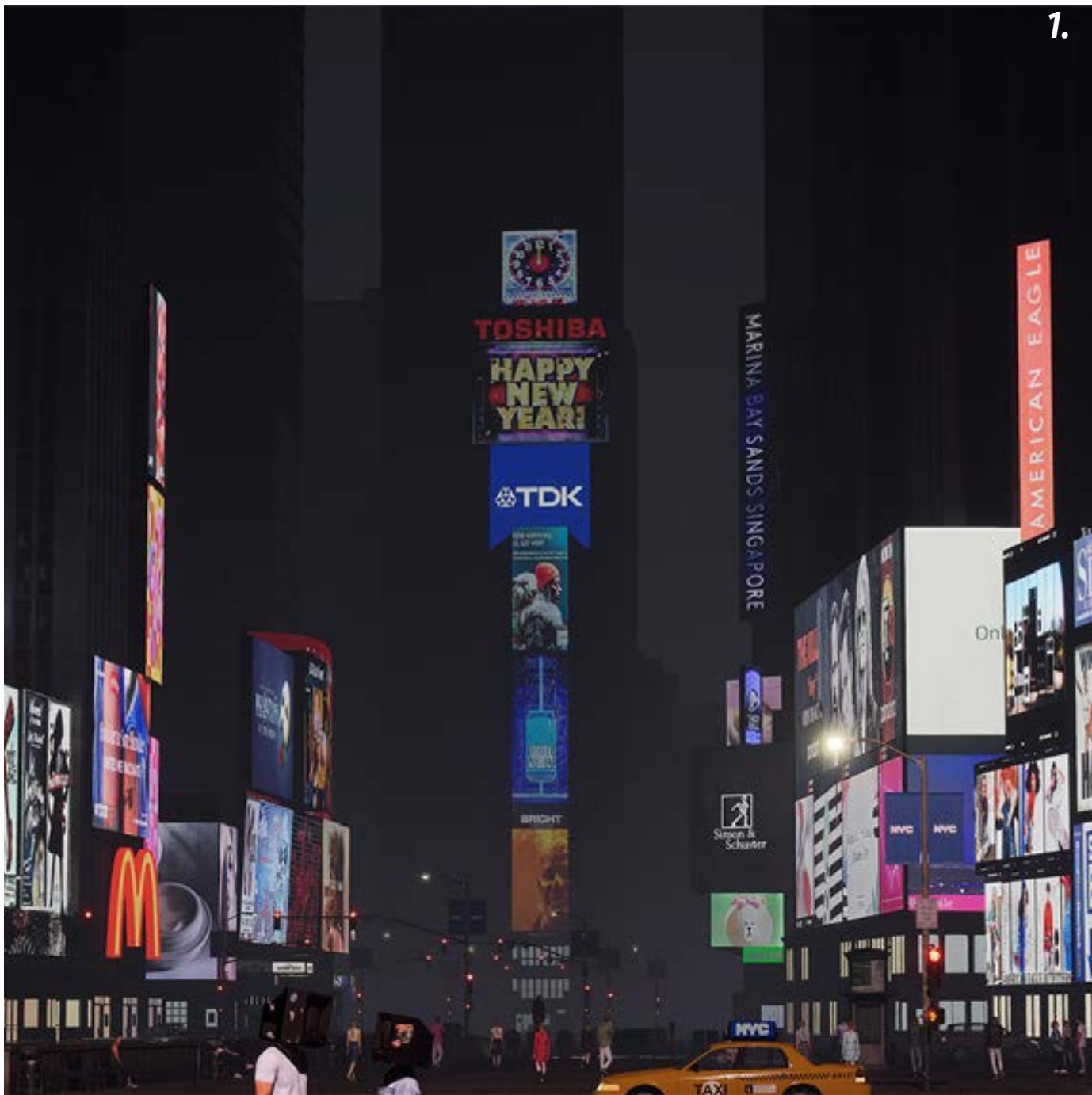
Evolution of the Urban Façade

Originally known as Longacre Square, Times Square developed alongside the growth of theatre, illuminated signage and commercial media. Across the twentieth century, the billboard evolved from an architectural accessory into an increasingly autonomous media surface, shaped by periods of theatrical spectacle, wartime blackout, commercial redevelopment and the later expansion of LED display systems.

The drawing traces this transformation across the surrounding façades, showing how successive layers of signage gradually overtook the architectural identity of the buildings themselves. By the digital era, the billboard no longer simply occupied the façade—it had effectively become the façade.

**Media Breaks Free –
From Façade to Spatial Structure**

As media progressively overtook the façades of Times Square, the project proposes a reversal: billboard surfaces are detached from the buildings and concentrated into a single mobile media sphere, allowing the original architecture to re-emerge. Once released from the façade, the billboard is no longer treated as an applied surface but as an inhabitable spatial structure.



1. Times Square Today

Architecture is increasingly concealed behind billboard infrastructure and digital media surfaces.

2. Media Concentration

Billboards are detached from surrounding buildings and reassembled into a mobile media sphere.

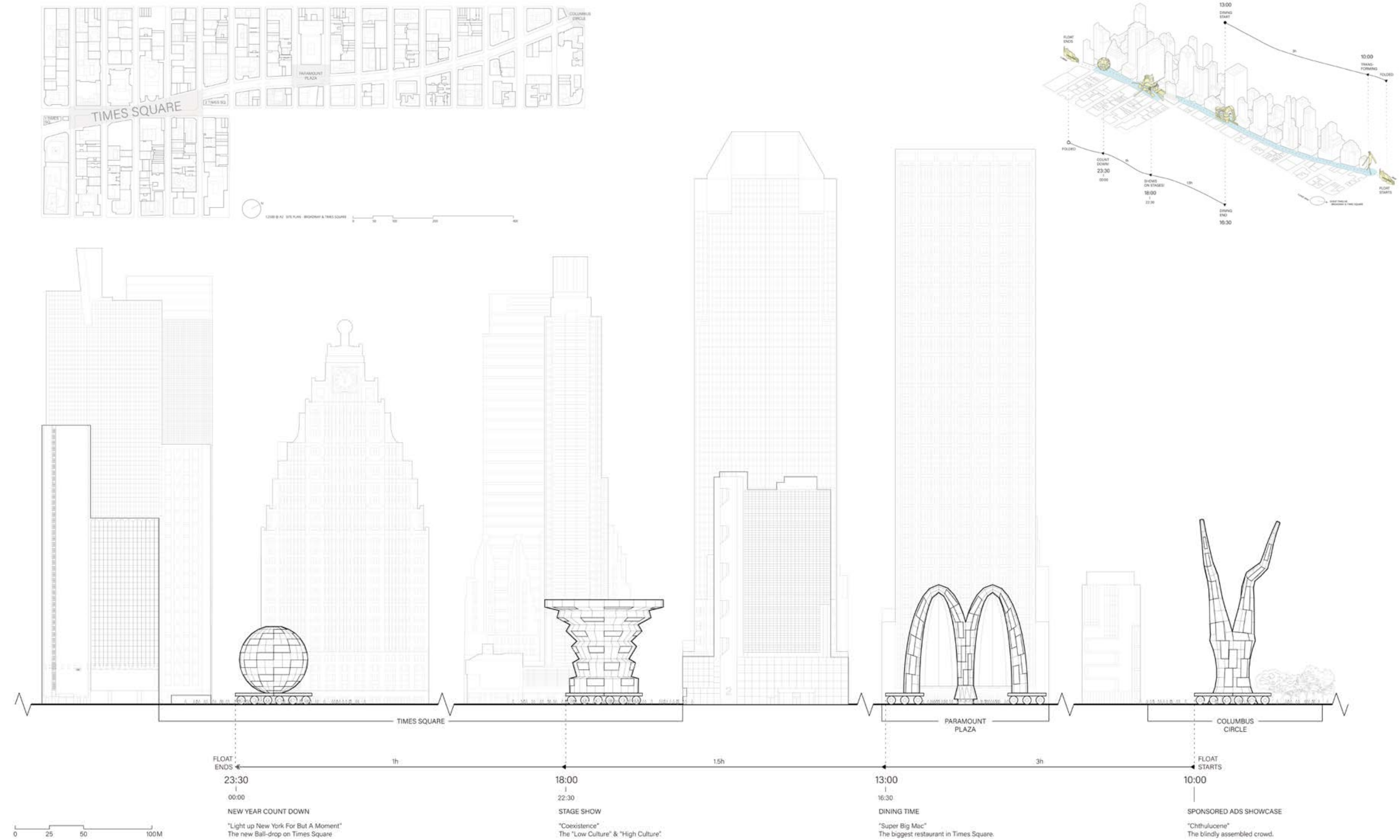
3. Media as Architecture

Freed from the façade, the billboard acquires structure, depth and interior space.



Media Takes Form

The project reimagines the New Year's Eve Ball Drop as a temporary urban ritual rather than a permanent media condition. Throughout the day, the mobile media structure travels along Broadway and shifts between different spatial, commercial and performative identities before its final transformation into the New Year's Eve ball. By concentrating advertising into a temporary moving structure, the surrounding façades are released from their permanent role as media supports.



**Transformation Sequence –
Unreal Engine 5 Visualisation**

The sequence was developed in Unreal Engine 5 to visualise four evolving identities, translating a moving architectural proposal into a cinematic urban experience.



1. Chthulucene

By 10 AM, the structure enters the city as an unfamiliar media spectacle, concentrating dispersed advertising into a single moving object.

2. Big-Mac

By noon, the structure reforms into the Golden Arches, temporarily becoming a monumental commercial architecture.

3. Coexistence

At 6 PM, the structure opens into an urban stage, shifting from commercial media to collective performance.

4. Light Up New York For But A Moment

At midnight, the structure transforms into the New Year's Eve ball and completes its journey towards One Times Square.



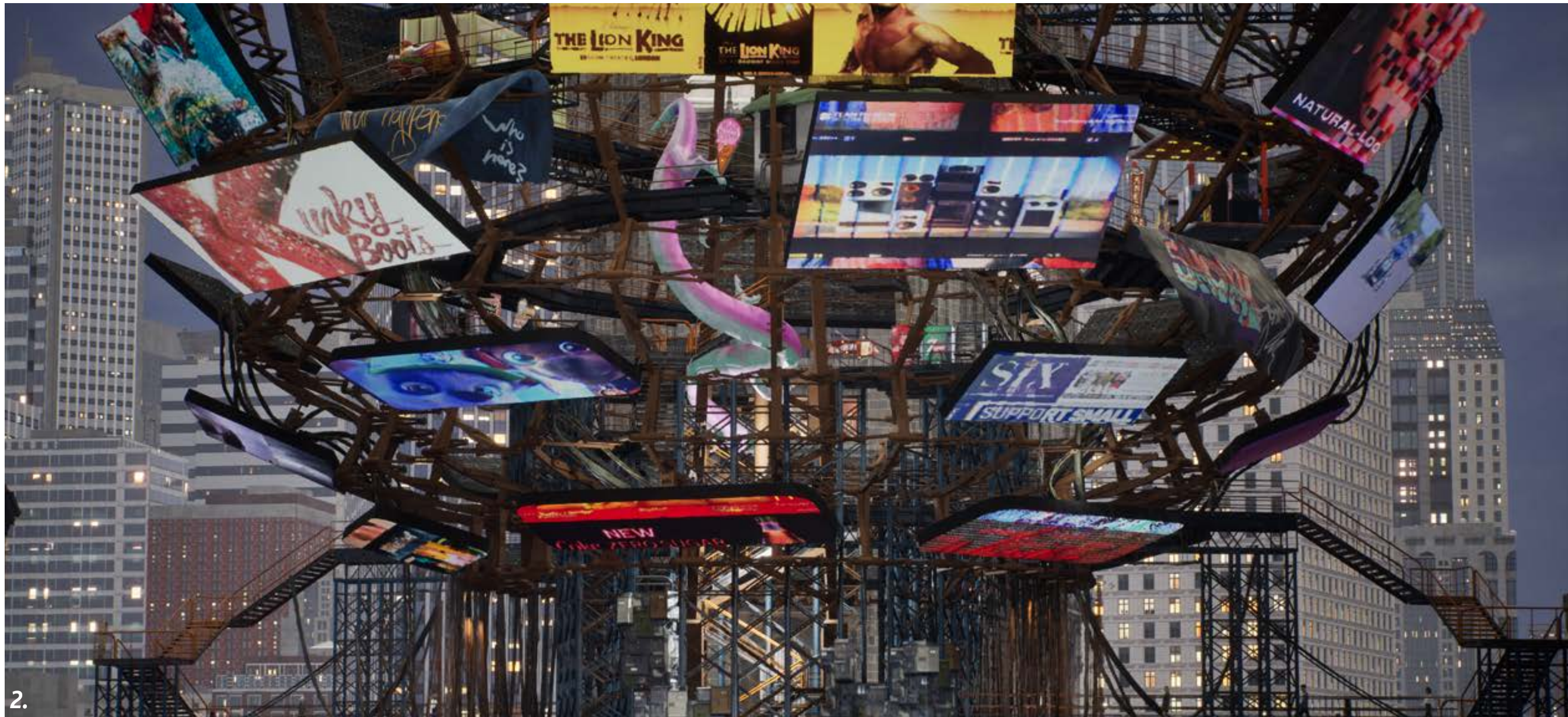
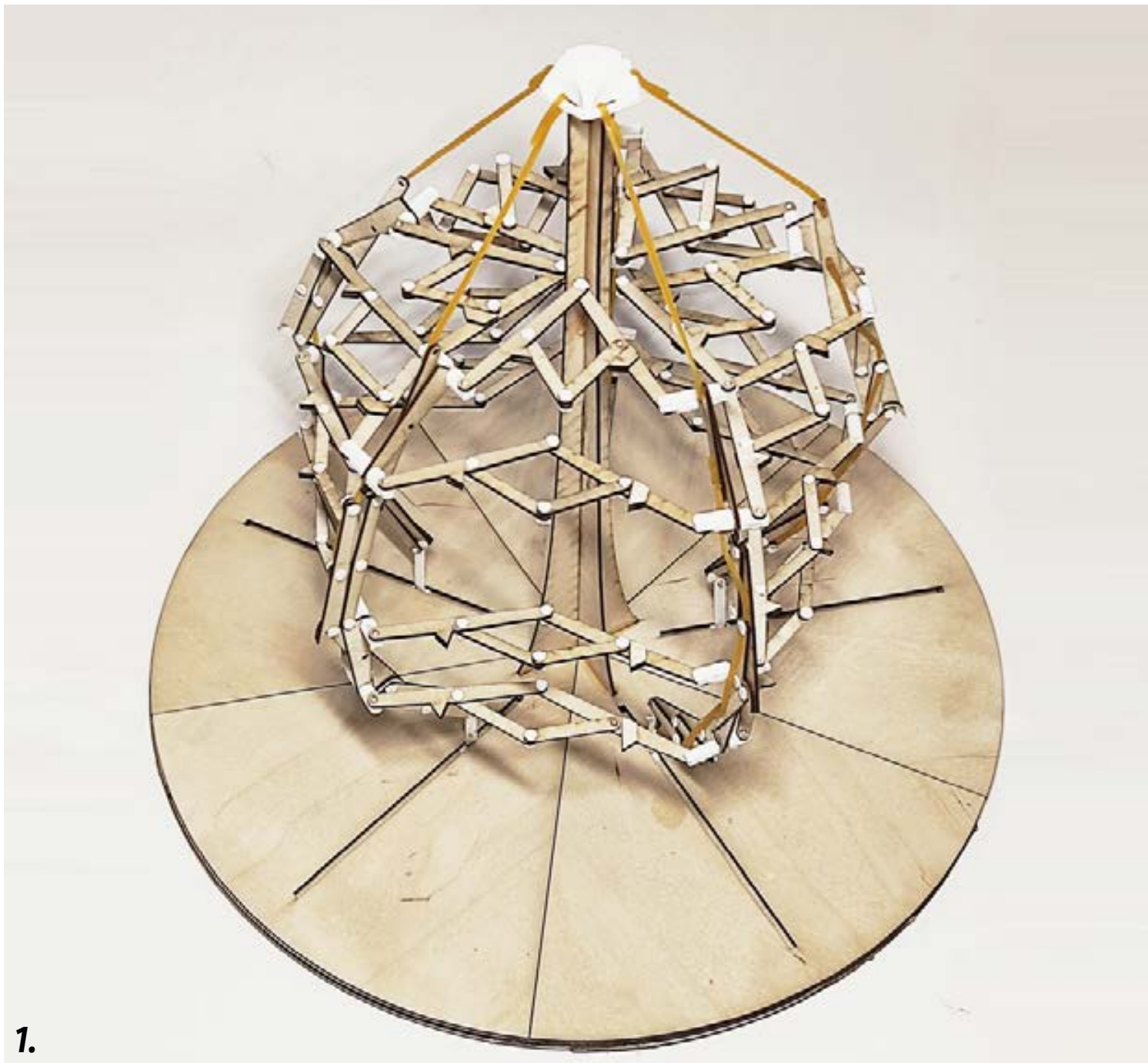
Final film – 8 mins



Structure, Movement, Form

Kinetic Prototyping & Structural Iteration

A series of physical and structural studies tested how a single media structure could shift between different configurations. A kinetic prototype explored movement and mechanical response, while the resulting transformation logic was translated into an extended structural study across the different stages of the parade.



1. Kinetic Prototype

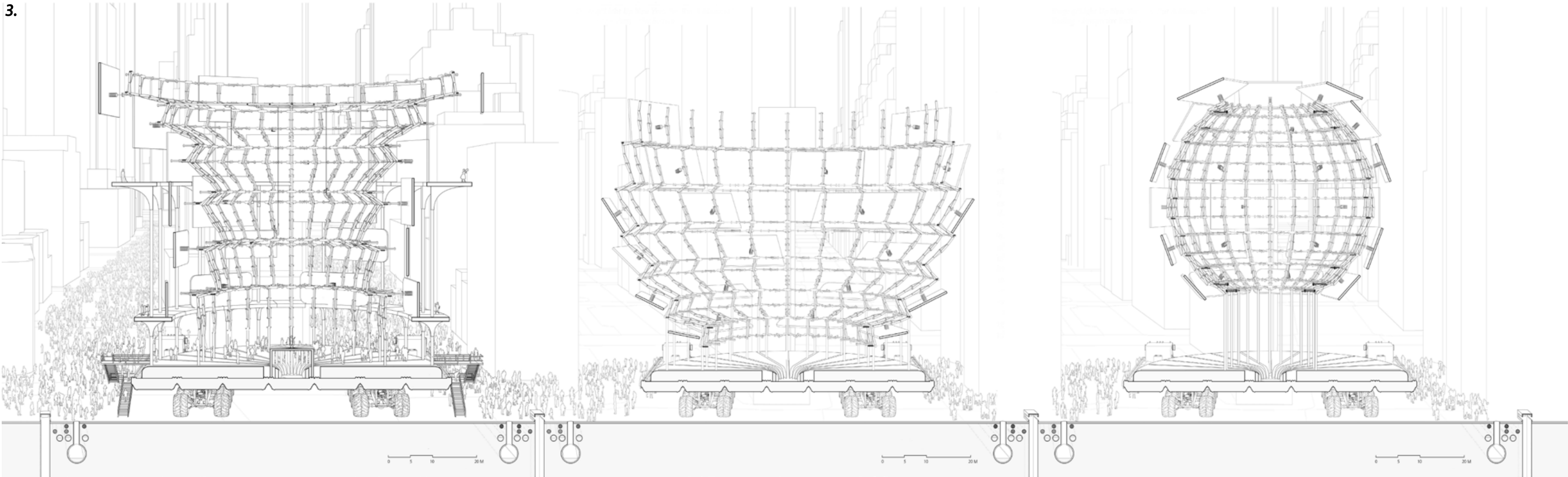
Physical testing of the structural transformation between configurations.

2. Media Sphere

Dispersed media surfaces are concentrated into a single inhabitable architectural structure.

3. Structural Transformation Section

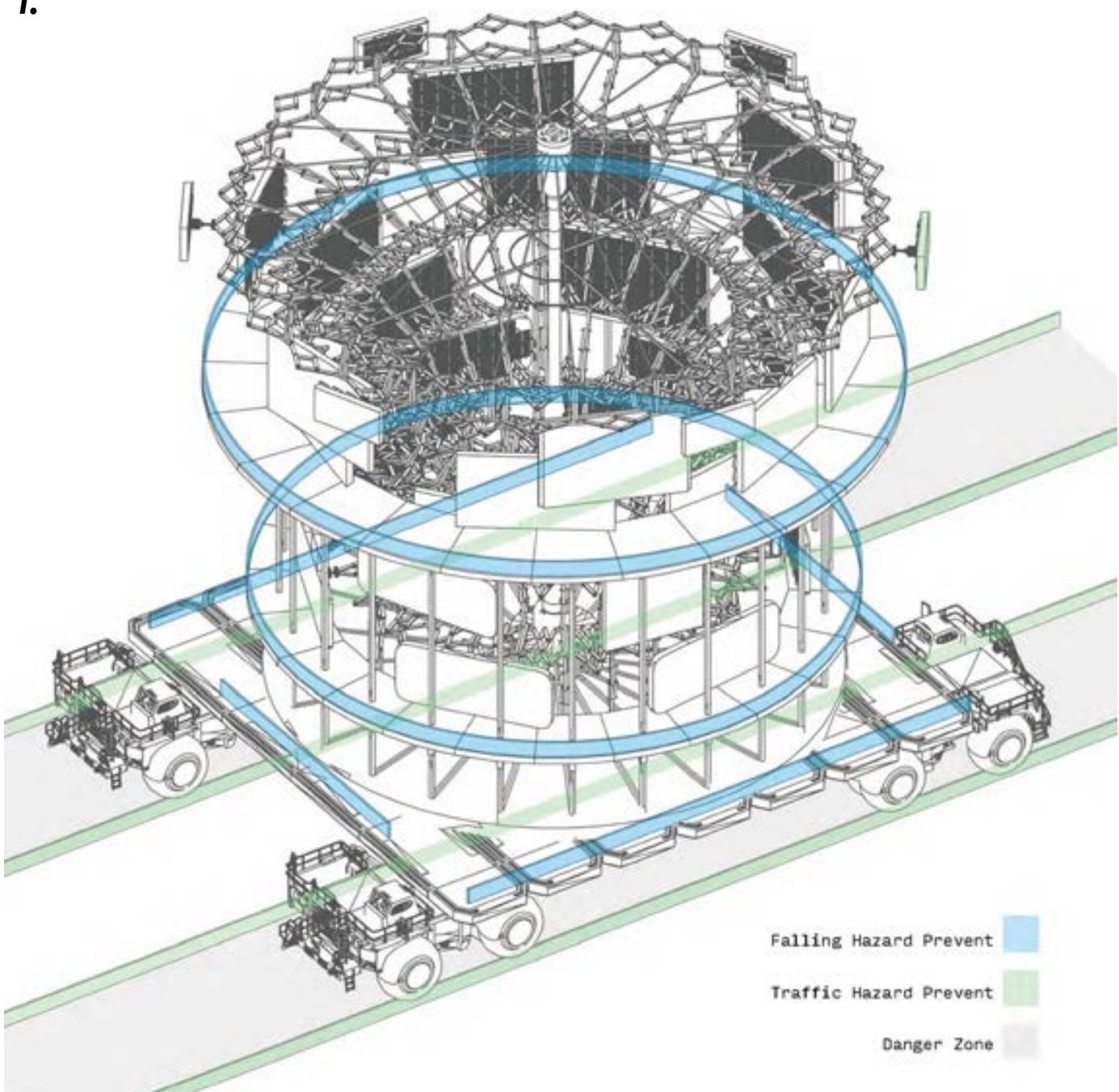
An extended sectional study showing how a single structural system reconfigures across the different stages of the parade.



The Billboard Becomes Architecture

The final transformation is no longer understood as a billboard enlarged to the scale of a building. Screens become enclosure, supporting frames become structure, and the volume once occupied only by advertising becomes inhabitable space. By separating media from the existing façades, the project produces two architectures at once: the buildings of Times Square regain their identity, while the billboard gains one of its own.

1.



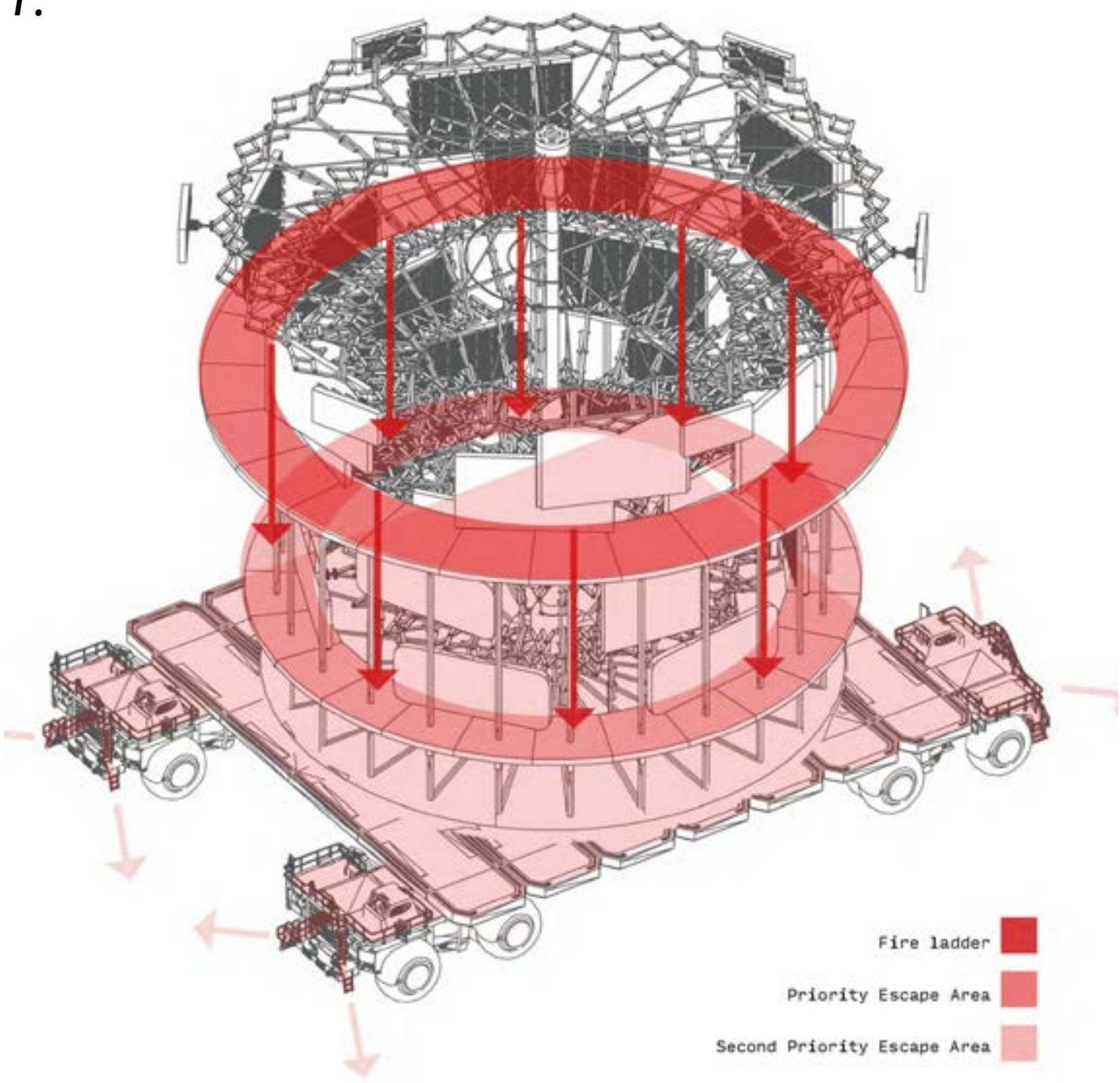
1. Technical Isometric

The mobile structure integrates circulation, inhabitable platforms and media surfaces within a single transformable frame. The diagrams define movement zones, access routes and spatial organisation within the architectural assembly.

2. Inside the Billboard

Advertising, once experienced only as an exterior image, becomes an interior condition that can be entered, occupied and experienced.

1'.



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2.



3. NO CURTAIN CALL

IMMERSIVE THEATRE · REAL-TIME INTERACTION · VR

No Curtain Call was developed through a year-long collaboration with theatre artists and community workshops in Archway, combining physical performance, VR, real-time interaction and spatial design. The project evolved from a VR experiment into a public performance and installation, linking physical stage elements with their digital counterparts.

1. Physical–Virtual Journey - Tactile VR Environment

The game map traces the transition from the physical site into the virtual world. Reclaimed bed frames are transformed into movable walls, giving the immersive VR journey a tactile architectural counterpart: while the virtual environment continuously changes, physical boundaries remain present to be touched, moved and navigated.

2. AprilTag Object Tracking - Digital Twin Interaction

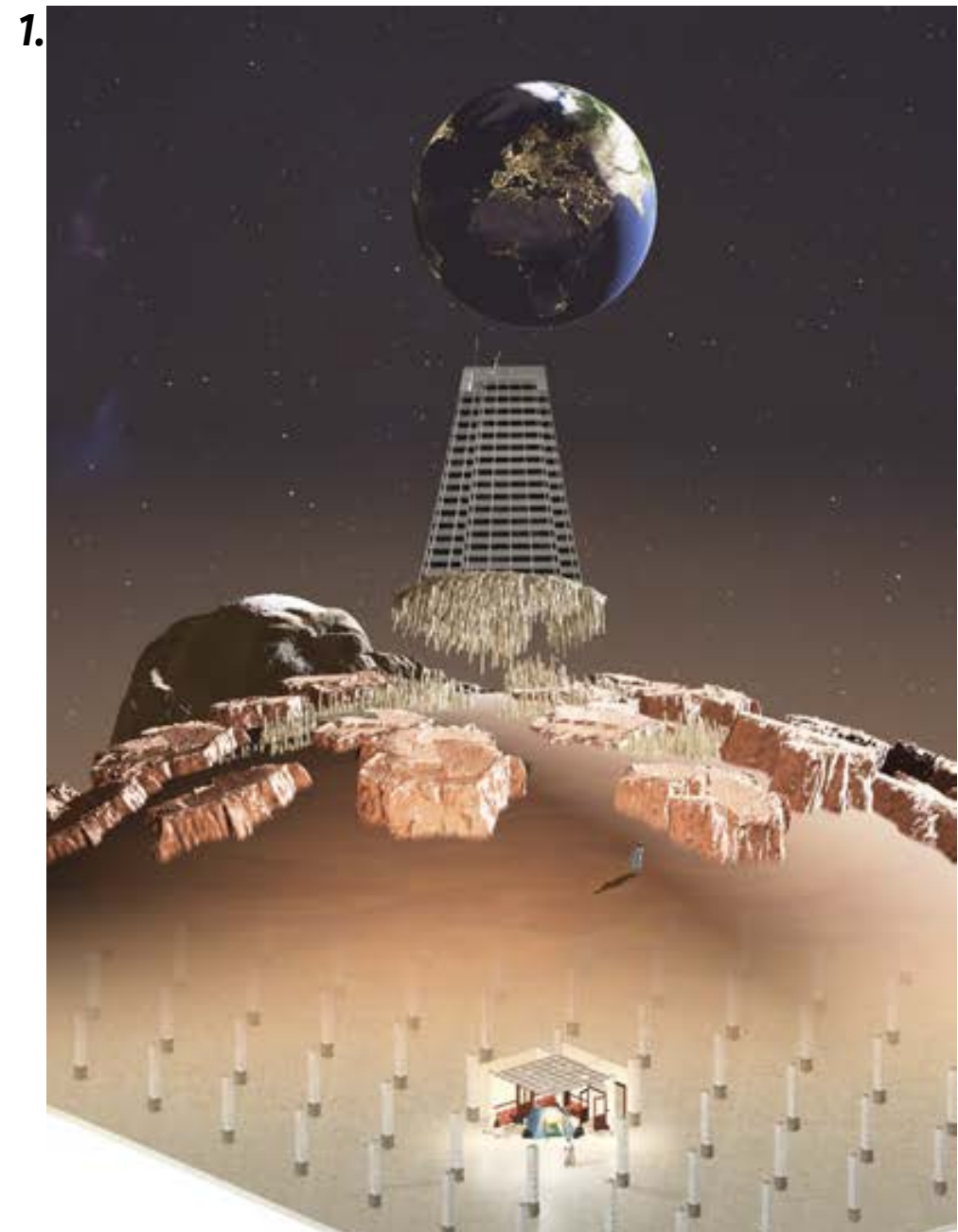
Physical objects and stage elements are tracked through a custom AprilTag system, allowing their movement to be mapped directly into the virtual environment.

3. Character Digitisation – Physical–Virtual Performance

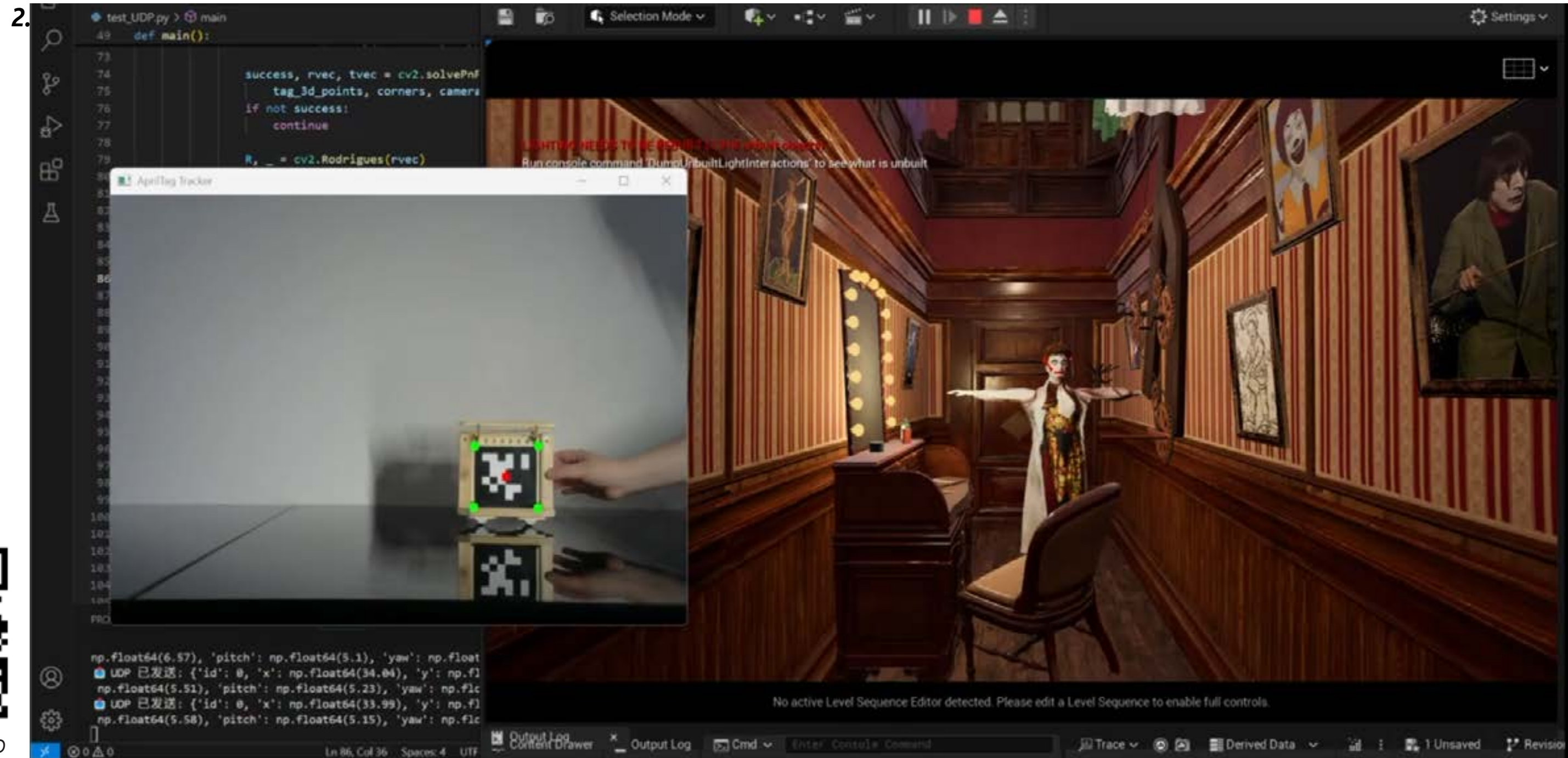
The clown performer was digitally reconstructed as a virtual character, allowing the theatrical identity to continue across physical and virtual environments.

4. Inertial Motion Capture – Real-Time Performance

Performer movement is captured and transferred into Unreal Engine in real time, connecting physical performance directly with the virtual scene.



Testing Video



VR Performance

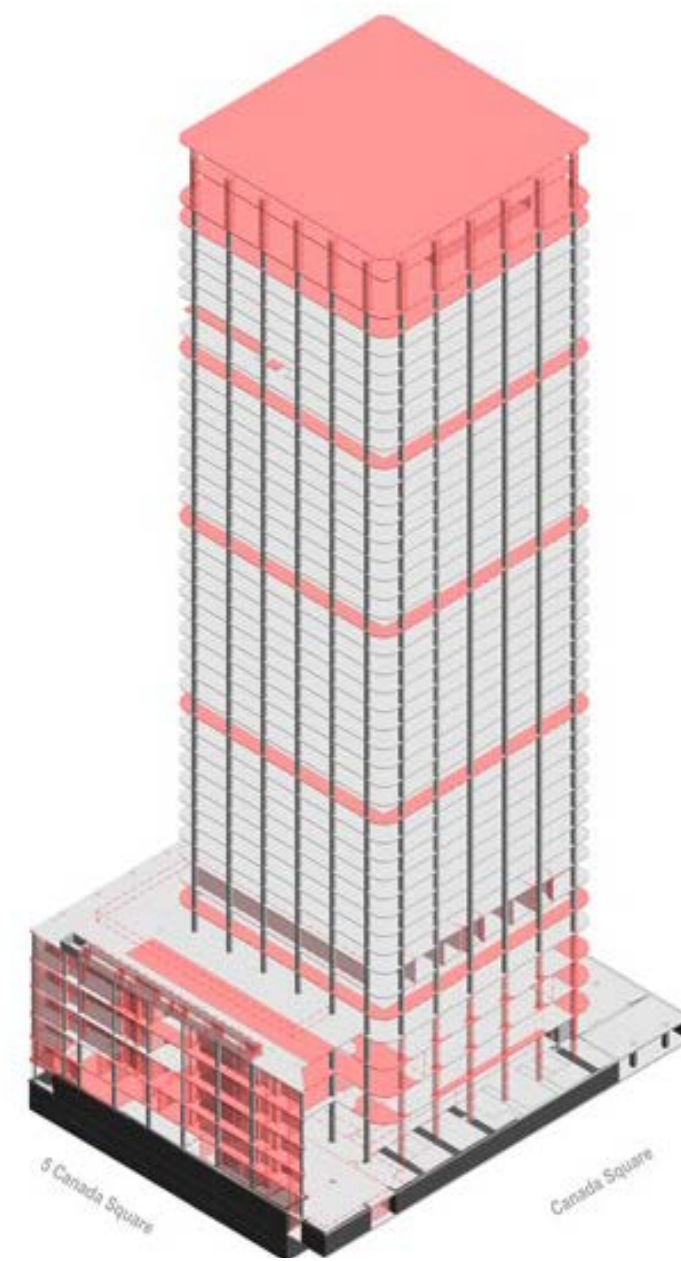


4. PROFESSIONAL PRACTICE

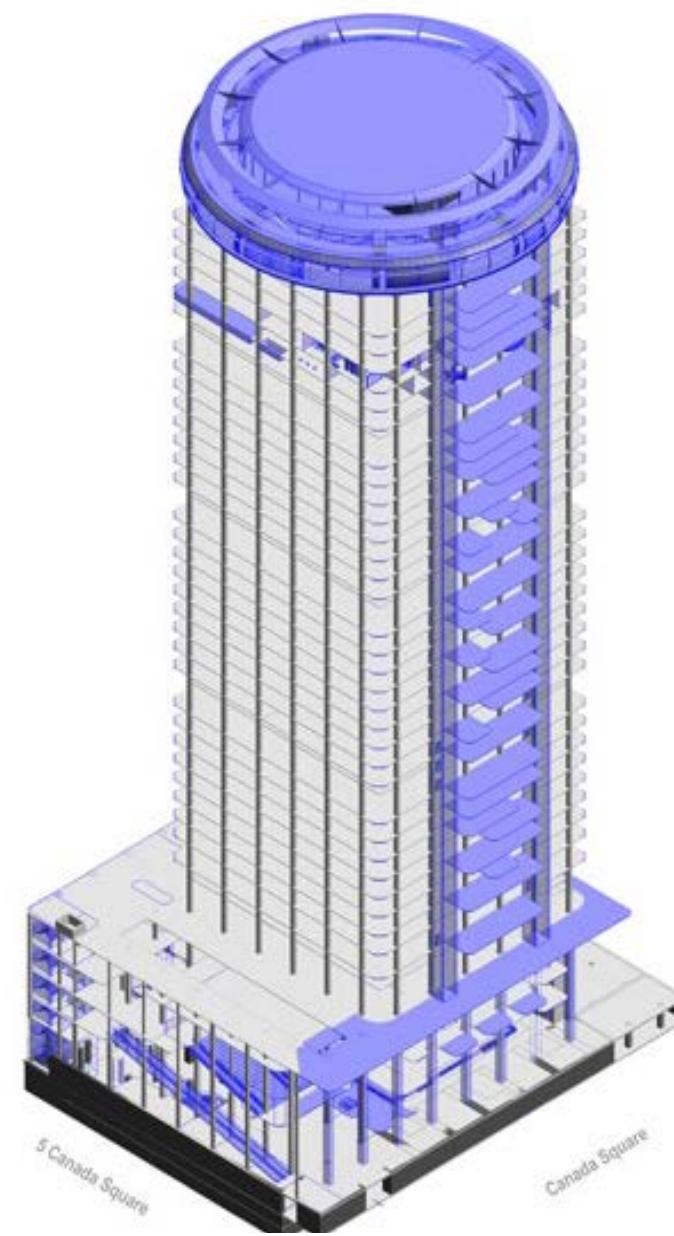
PROFESSIONAL EXPERIENCE AT
FOSTER + PARTNERS · STUDIO 1

8 CANADA SQUARE

MODEL COORDINATION & REAL-TIME VISUALISATION



SW -
Existing+Demolition



SW -
Retained+Proposed



Tower Top - Observation Deck I

Developed as a competition proposal, 8 Canada Square involved the transformation of the existing tower through selective retention and new intervention. I took primary responsibility for the Revit model and cinematic visualisation, coordinating the architectural model and translating it into a real-time Unreal Engine environment.

The Tower Top exhibition space became a key focus of the visualisation workflow, using real-time spatial storytelling to communicate the proposal as an inhabitable public destination for exhibition, observation and cultural experience.



Tower Top - Observation Deck II

1 SPITALFIELDS

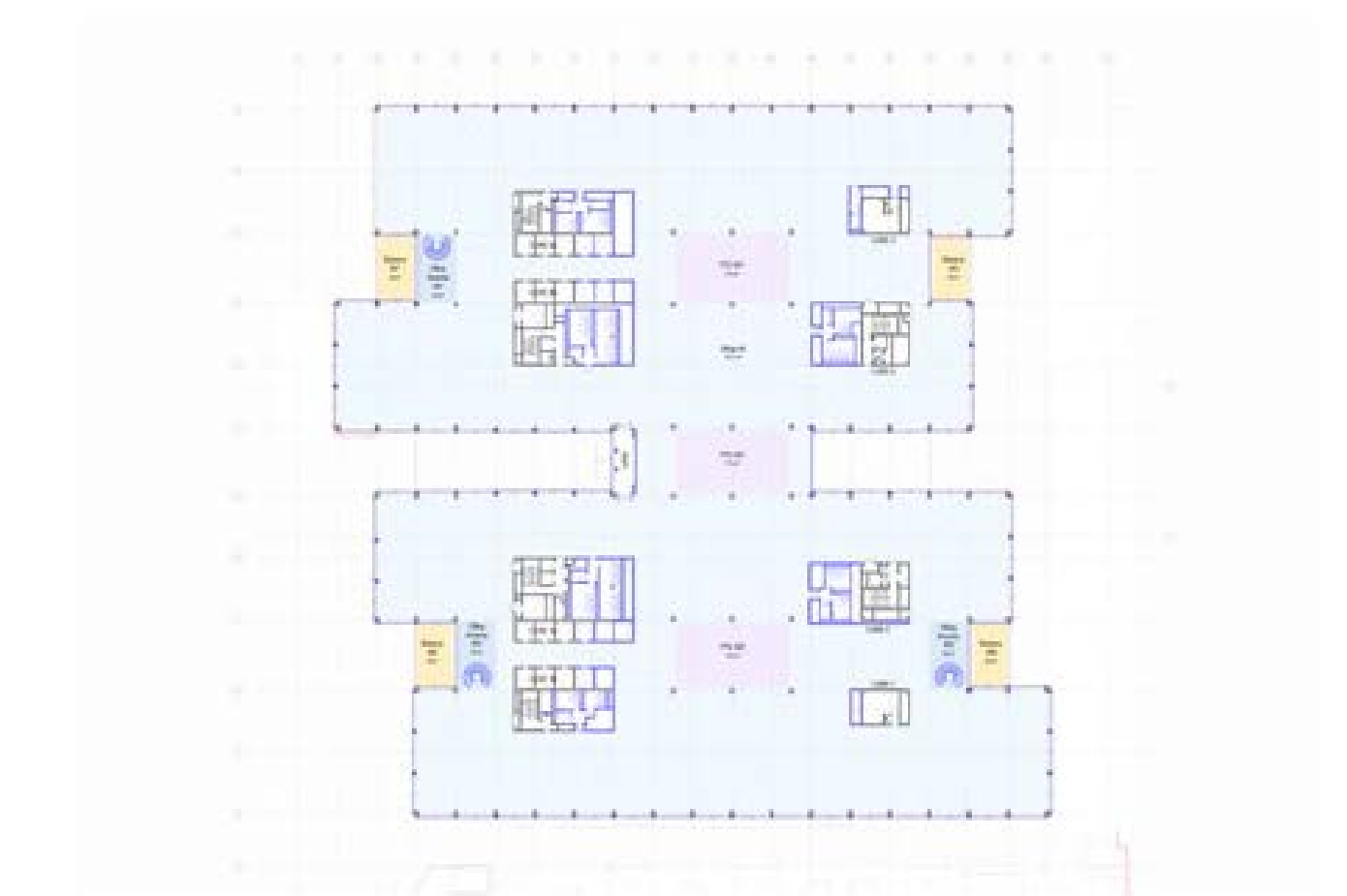
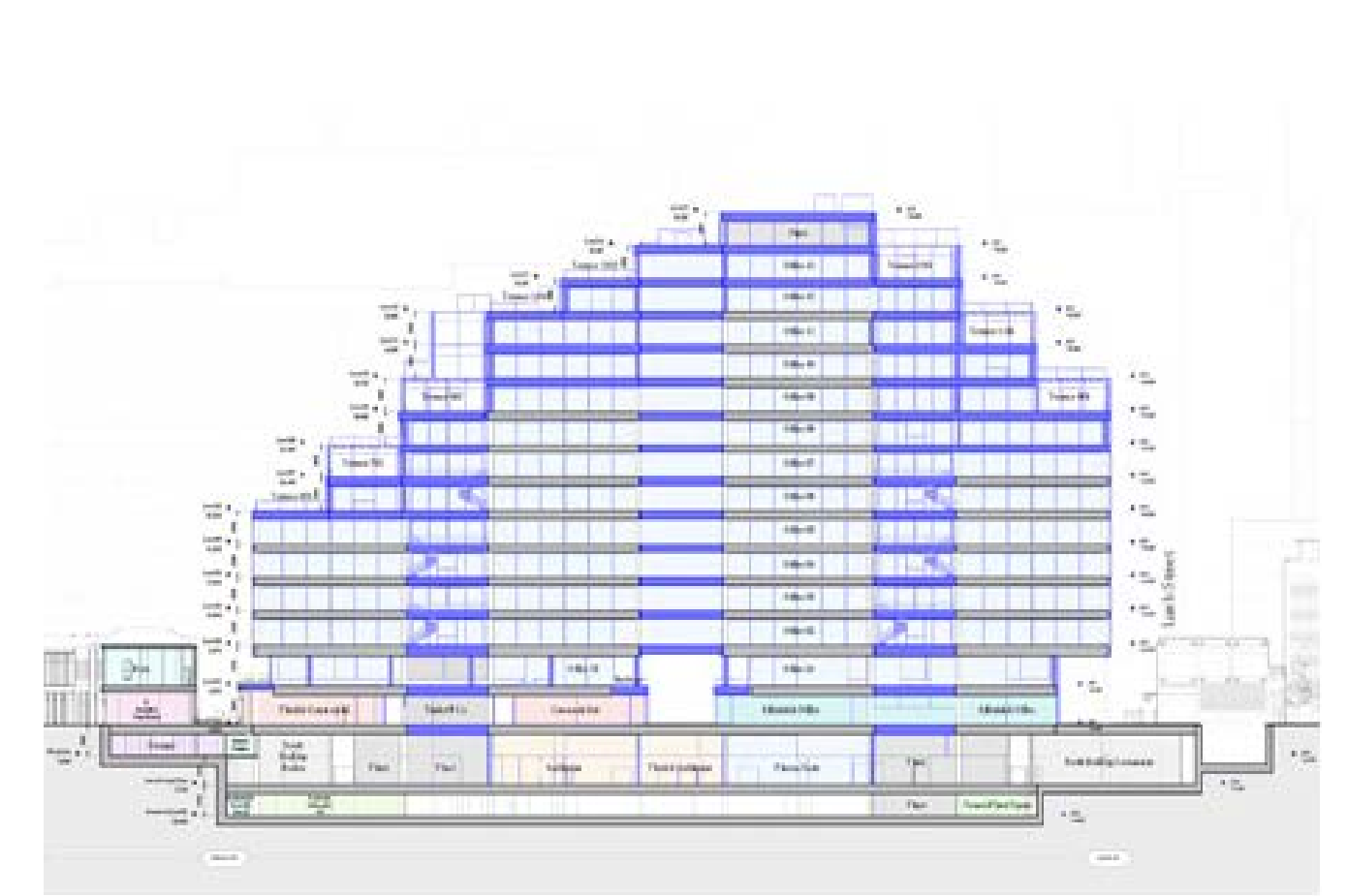
MODEL COORDINATION & VISUAL COMMUNICATION

I worked on 1 Spitalfields through its transition from RIBA Stage 2 to Stage 3, contributing to architectural design, Revit coordination and visualisation across the main building, market and affordable workspace.

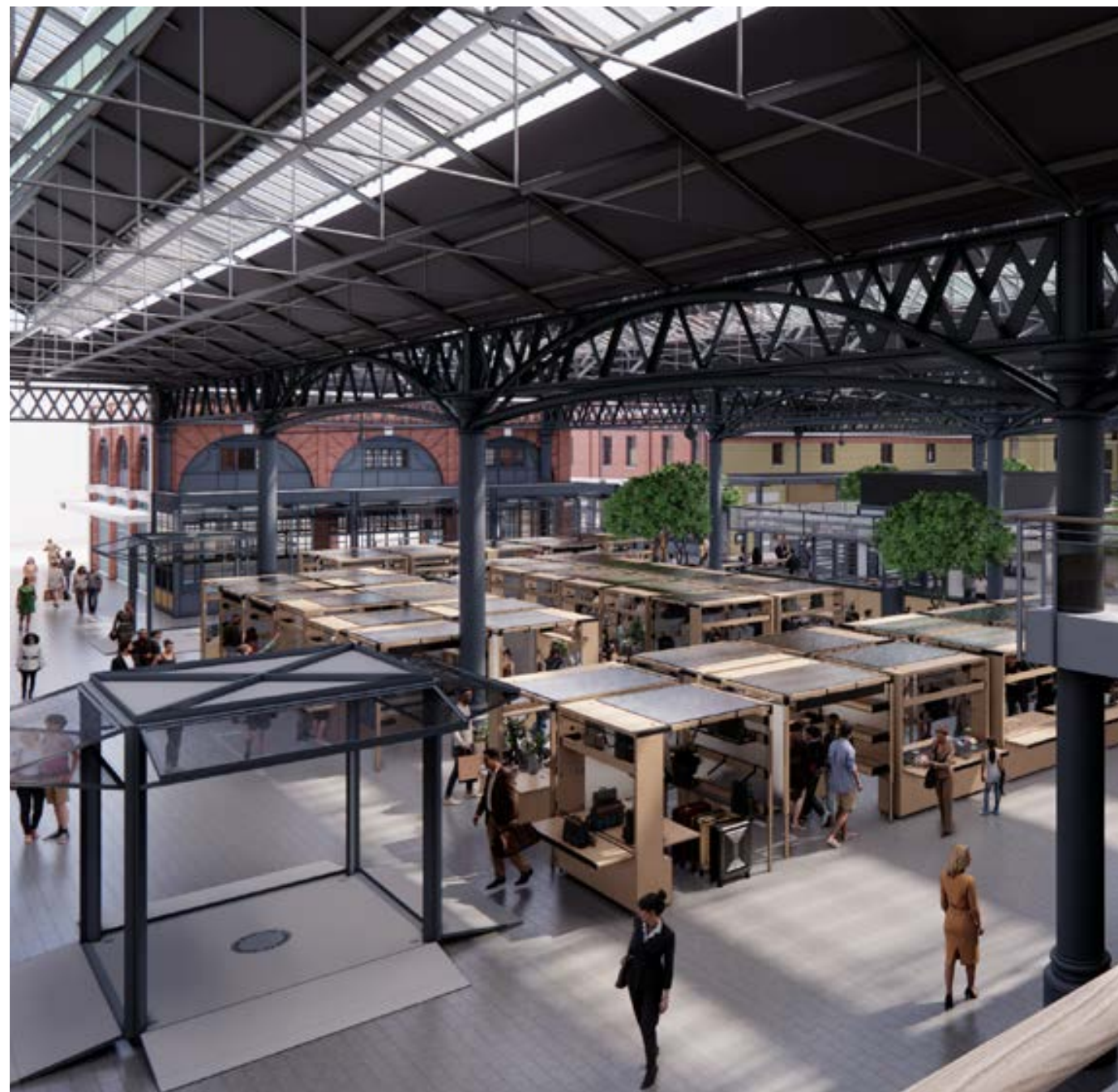
My role included producing and coordinating general arrangement and detailed drawing packages across existing, demolition and proposed conditions, while translating coordinated architectural information into clear visual outputs.



Main Building - Proposed Perspective



Main Building - Proposed Revit Drawings



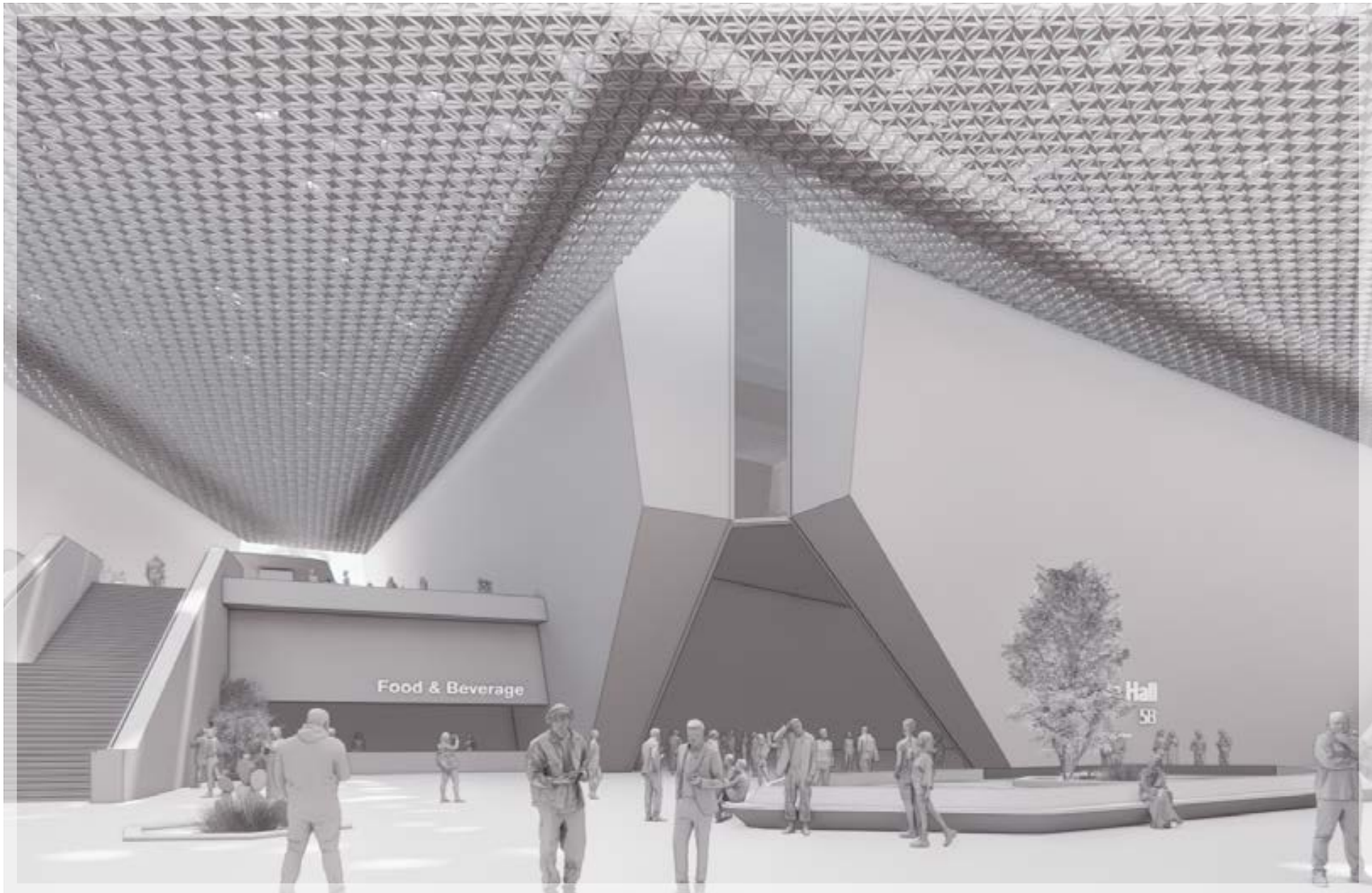
Market - Proposed View I



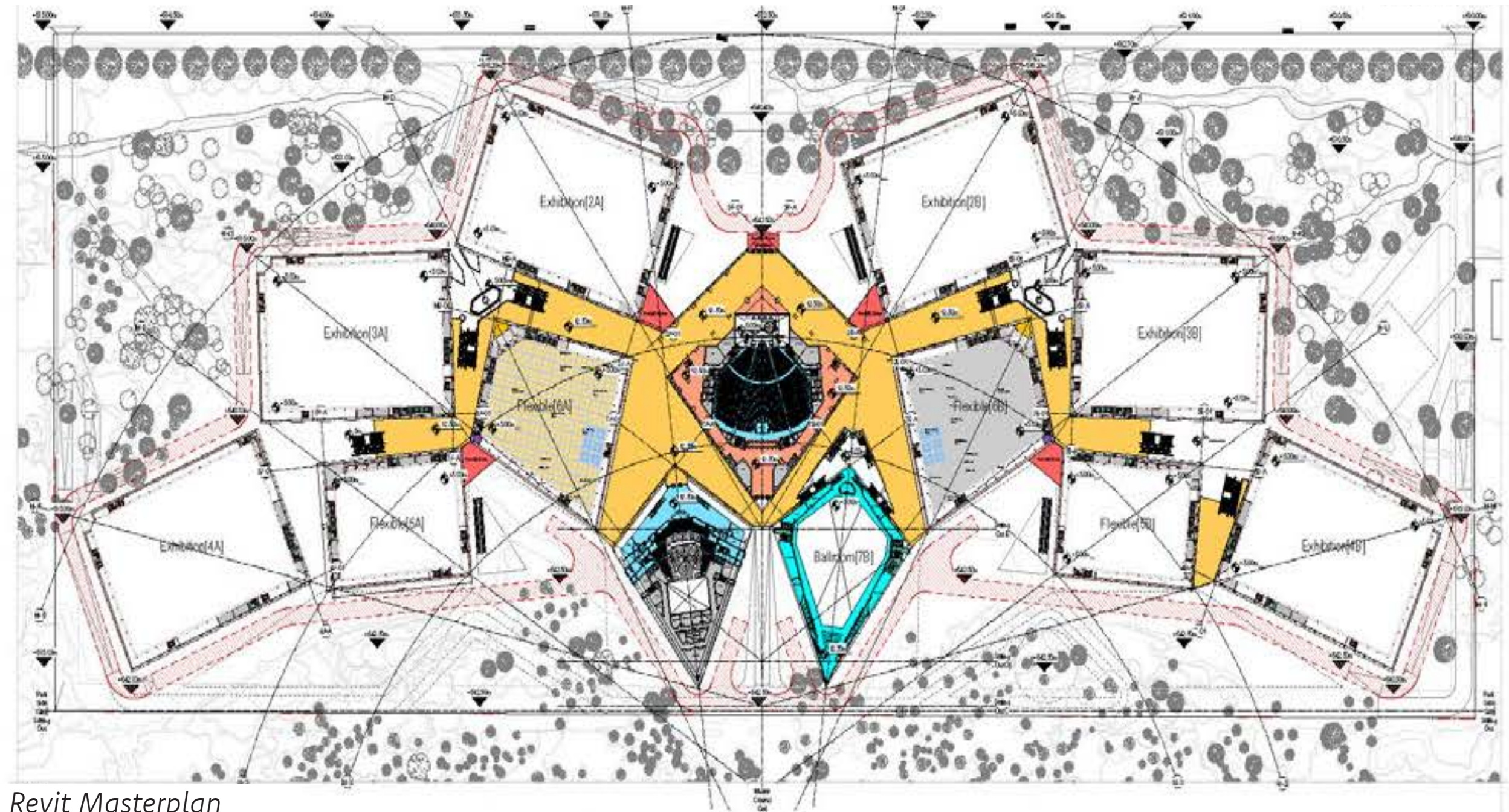
Market - Proposed View II

RECC

PARAMETRIC DESIGN · MODEL COORDINATION ·
VISUALISATION



Main Entrance



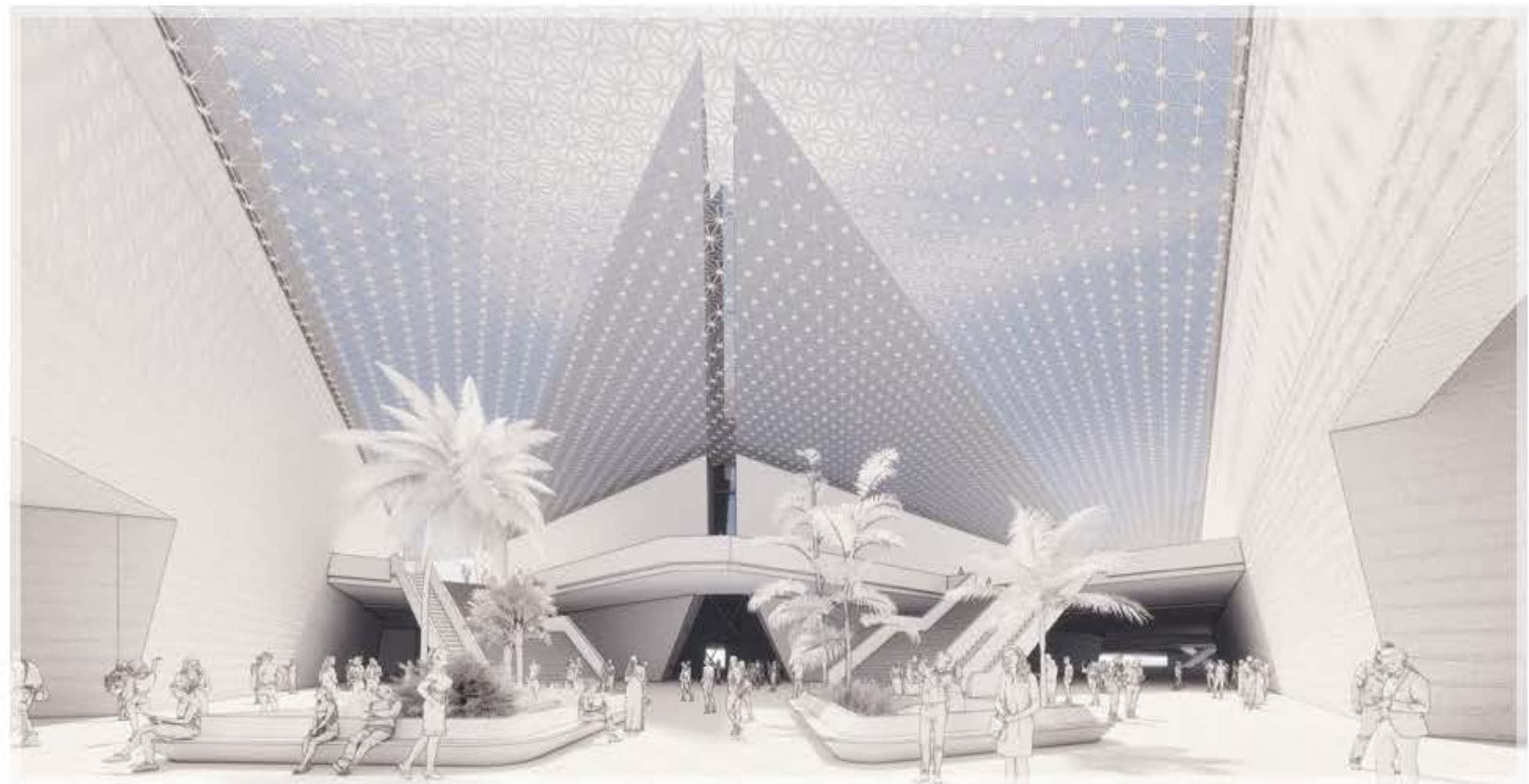
Revit Masterplan

Developed at RIBA Stage 2 within a multidisciplinary team of more than twenty, RECC required continuous model coordination and visual development across a large exhibition complex. My role included architectural design development, Revit coordination and visualisation across the entrance, mezzanine canopy and sunken communal areas.

I also contributed to a Grasshopper-based parametric shading system, testing geometric variations as part of the wider façade strategy.



Mezzanine Area



Sunken Communal Area